Supplementary Material
MMSE Pair Scheduling: Sharp Complexity, Certified Power Allocation, and Regime-Dependent Rate Gains

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Sections S1–S4 give the channel construction, curvature coefficients, relaxation examples and full structural proofs. Sections S5–S6 describe the reference methods, verification and ray-traced setup/results; Sections S7–S9 give the power-component adaptations, controlled-array comparison and fixed-assignment and small-network references. Section S10 explains reproduction and identifies the archived studies. Section S11 develops the finite-box/integral scheduling bound and reports its full comparison on the same 32 ray-traced networks. Section S12 documents the ray-traced receive-dimension cross-check and the exact-search details. Section S13 documents the standardized 3GPP CDL benchmark used for the rate claim. The constructed hardness channels, ray-traced channels and controlled arrays are distinct input families.

S1 Rational channel construction

Theorem 3.16 of the AIM Minimum Rank Group [1], p. 1639, gives a faithful orthogonal representation of the complement of a tree in $\mathbb{R}^3$, with distinct representing vectors linearly independent. We extend this construction to rational unit vectors drawn from a fixed finite alphabet for forests with at most seven roles. This additional restriction is required by Theorem 1 of the main article.

Root each component of a forest with at most seven roles and visit parents before children. The first root receives $\mathbf{e}_3$. A subsequent component root is chosen from

$$\mathbf{r}(t) = \frac{(2t, 2t^2, 1 - t^2 - t^4)^T}{1 + t^2 + t^4}, \qquad t \in \{0, \dots, 100\}. \tag{A1}$$

These are rational unit vectors. For any previous nonzero vector $\mathbf{w}$, the numerator of $\mathbf{w}^T\mathbf{r}(t)$ is a nonzero polynomial of degree at most four. Orthogonality excludes at most four values of t. Parallelism of unit vectors is equivalent to squared inner product one; after clearing denominators this excludes at most eight values. That polynomial cannot be identically zero because the nonconstant continuous curve would otherwise lie in $\{\mathbf{w}, -\mathbf{w}\}$. At most six preceding vectors forbid at most 72 real values, so an allowed integer exists.

For a child with parent vector $\mathbf{p}$, use $\mathbf{Q} = \mathbf{I}$ if $\mathbf{p} = \mathbf{e}_3$; otherwise let

$$\mathbf{Q} = \mathbf{I} - \frac{(\mathbf{e}_3 - \mathbf{p})(\mathbf{e}_3 - \mathbf{p})^T}{1 - p_3}. \tag{A2}$$

This rational Householder matrix is orthogonal and maps $\mathbf{e}_3$ to $\mathbf{p}$. Choose the child's vector from

$$\mathbf{c}_{\mathbf{p}}(t) = \mathbf{Q}\left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2}, 0\right)^T. \tag{A3}$$

It is unit and perpendicular to its parent. Previous nonparent vectors are nonparallel to $\mathbf{p}$, so undesired orthogonality excludes at most two values and parallelism at most four. An integer in the same candidate set

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remains available. In parent-before-child order, the parent is the only previous neighbor. Induction therefore realizes exactly the forest's zero-inner-product graph while keeping distinct roles nonparallel.

For a precise fixed alphabet, enumerate all labelled forests on $\{1, \dots, m\}$ for $1 \le m \le 7$. Root each component at its smallest label, order components by their roots, and traverse each in depth-first order with children in increasing label order. At each choice take the least admissible $t \in \{0, \dots, 100\}$. Define $\mathcal{A}$ as the union of all vectors output by (A1)–(A3), including the initial $\mathbf{e}_3$, over this finite enumeration. This specifies the alphabet exactly without an instance-dependent choice. Its cardinality and rational bit lengths are constants independent of K; a lexicographic ordering of its rational coordinates fixes the input indices. The construction is allowed to use the general forest family, although the reduction needs only forests with at most three edges and one outsider role.

S2 Coefficients for the curvature proof

For $N = \sum_{i,j} c_{ij} x^i y^j$ in (D5), Section S4, symmetry gives $c_{ij} = c_{ji}$. The nonzero coefficients for $i \ge j$ are

(i, j)	c_{ij}
$4, 4$	a^4
$4, 3$	$4a^4$
$4, 2$	$2a^2(a^2 + 4a - 2)$
$4, 1$	$4a(a^2 + a - 1)$
$4, 0$	$2a^2 - 1$
$3, 3$	$4a^2(4a^2 - a + 1)$
$3, 2$	$4a(2a^3 + 6a^2 - 3a + 1)$
$3, 1$	$4a(3a^2 + 3a - 2)$
$3, 0$	$4(a^2 + a - 1)$
$2, 2$	$2(2a^4 + 16a^3 - a^2 + 1)$
$2, 1$	$4(a + 1)(2a^2 + 2a - 1)$
$2, 0$	$2(-a^2 + 8a - 4)$
$1, 1$	$4(7a - 3)$
$1, 0$	$4(2 - a)(2a - 1)$
$0, 0$	$(3 - 2a)(2a - 1)$

For $a \in [1/\sqrt{2}, 1]$, $a^2 + a - 1 \ge a - 1/2 > 0$, $a^2 + 4a - 2 \ge 4a - 3/2 > 0$, $4a^2 - a + 1 > 0$, $6a^2 - 3a \ge 0$, $3a^2 + 3a - 2 \ge 3a - 1/2 > 0$, and $-a^2 + 8a - 4 \ge 8/\sqrt{2} - 5 > 0$. The remaining terms are nonnegative, while the constant coefficient is strictly positive. Thus the determinant numerator is positive throughout the stated domain.

S3 Assignment and power gaps in the configuration relaxation

The following constructed examples show how assignment mixing and power mixing can each prevent a configuration-LP upper bound from being attained by a feasible single-slot schedule. General-graph matching and frequency-sharing convexification provide the relevant background [2, 3].

S3.1 Assignment mixing with channel-realizable weights

Take six users, three resource-invariant receive dimensions and unit powers. The first three channels are the standard basis of $\mathbb{R}^3$; the other three are the columns of

$$Q = \frac{1}{7} \begin{pmatrix} 6 & -2 & -3 \\ -2 & 3 & -6 \\ -3 & -6 & -2 \end{pmatrix}.$$

Writing $v = (1, 2, 3)^T$ gives $Q = I - vv^T/7$ and $Q^T Q = I$. Each group of three is internally orthogonal, whereas all cross-group correlations are nonzero. Their smallest squared correlation is $4/49$. At unit

normalized powers the pair weight is

$$w(r) = \ln \frac{(4-r)^2}{4}.$$

Every perfect matching has either one or three cross-group edges. A matching containing one cross edge with $r = 4/49$ and two within-group edges is optimal and has value

$$V_{\mathrm{IP}} = 2\ln 4 + \ln \frac{9216}{2401}.$$

There are six within-group edges, three in each triangle. Assign configuration mass $1/6$ to each such edge on each of the three resources. Every user then has total incidence one and every resource has total mass one. This LP solution has value $3\ln 4$, which is optimal because no pair exceeds $\ln 4$. Consequently,

$$V_{\mathrm{LP}} - V_{\mathrm{IP}} = \ln \frac{2401}{2304} = 2\ln \frac{49}{48} > 0.$$

This is a finite-grid example with a singleton unit-power grid and total budget six. It also explains the failure of an unrestricted permutation assignment to enforce disjoint pairs: the two directed three-cycles attain twice the fractional value but are not a pairing. The homogeneous integer problem is still a polynomially solvable general-graph matching problem. The example does not concern formulations that intentionally permit cyclic reuse of users.

S3.2 Power mixing after the pair is already fixed

Take one resource, two unit-gain collinear channels, powers in $[1, 2]$, and total budget $11/5$. Here

$$q(x, y) = \ln \frac{(1+x+y)^2}{(1+x)(1+y)}.$$

Both partial derivatives are nonnegative on this box: for example, $q_x = (1+x-y)/[(1+x+y)(1+x)]$. An optimum therefore exists on $x + y = 11/5$. On this line the numerator is constant and the denominator is concave in x; its minimum is at $x = 1$ or $x = 6/5$. Hence the integral continuous optimum is $\ln(128/55)$.

The continuous configuration LP can mix weight $4/5$ at $(1, 1)$ and $1/5$ at $(1, 2)$. Its average power is $11/5$ and its value exceeds the deterministic optimum because

$$\left(\frac{9}{4}\right)^4 \frac{8}{3} > \left(\frac{128}{55}\right)^5.$$

The decisive comparison uses rational arithmetic; no rounded logarithm is needed. The pairing is already fixed, so branching only on assignment variables cannot resolve this gap. A single-slot interpretation requires optimization of actual powers, whereas a time-sharing interpretation would describe a different feasible service model.

S4 Full proofs of the structural theorems

We give the full proofs of the structural theorems and the finite-tolerance refinement result. Equations (D1)–(D9) are supplementary; other numbered equations refer to the main article. Sections S1 and S2 provide the rational representation and determinant coefficients, respectively.

S4.1 Proof of Theorem 1

Proof. Use perfect three-dimensional matching with sets X, Y, Z of equal cardinality n, where every element occurs in at most three triples. Dyer and Frieze [4], Section 2, p. 141, statement (2.1), state NP-completeness for the planar perfect-matching variant in which every element occurs in two or three triples, and attribute its proof to their subsequent planar-3DM paper. Dropping planarity gives the premise needed here. The construction below realizes the reduction with finite-dimensional channel vectors. Create users $X \cup Y$,

resources Z, and $K = 2n$. For a fixed resource z, the admissible triples induce a bipartite graph F_z with at most three edges. It is a forest with at most six nonisolated vertices. Add an isolated outsider role.

Section S1 constructs a rational unit representation in $\mathbb{R}^3$ whose zero inner products are exactly the edges of this at-most-seven-role forest. Each nonisolated user receives its role vector on resource z; every remaining user receives the outsider vector. Two outsiders have inner product one, while an outsider and an active role have nonzero inner product. Hence a user pair is orthogonal on z exactly when it corresponds to a triple of the original instance.

At unit power, each selected pair has

$$q_{uvs} = 2\ln\left(2 - \frac{|\mathbf{h}_{us}^T \mathbf{h}_{vs}|^2}{2}\right) \le 2\ln 2, \tag{D1}$$

with equality exactly for an orthogonal pair. A full schedule attains $K\ln 2$ exactly when the original instance has a perfect three-dimensional matching. IDEAL-MMSE-PAIR encodes K, S as integers and its channel table as KS indices into the fixed ordered alphabet of Section S1. A certificate lists the two user indices for every resource, using $O(K \log K)$ bits; incidence and range checks enforce exactly-once service. To avoid transcendental comparison, test $\prod_i (1+\gamma_i) \ge 2^K$. Each factor is rational with constant bit length. Multiplication gives numerator and denominator with $O(K)$ bits, and cross-multiplication against the $(K+1)$-bit integer 2^K is polynomial-time. Malformed instances and invalid schedules are rejected. This proves NP-membership; the reduction proves NP-hardness. No arbitrary real threshold is part of this language.

Let $b = \min\{|v^T w| : v, w \in \mathcal{A},\ v^T w \neq 0\} > 0$, including $v = w$ for two users assigned the outsider vector. Any no instance loses at least $\delta_0 = -2\ln(1-b^2/4) > 0$ from the ideal sum. Under continuous powers, rewrite the pair rate as

$$\begin{aligned} q_{uvs} = {} & \ln(1+p_u) + \ln(1+p_v) \\ & + 2\ln\left(1 - r_{uvs}\frac{p_u p_v}{(1+p_u)(1+p_v)}\right). \end{aligned} \tag{D2}$$

Jensen's inequality bounds the sum of the first terms by $K\ln(1+\bar{p})$. Every no instance contains a forbidden pair, whose interaction loss is at least

$$\delta_{\mathrm{box}} = -2\ln\left(1 - b^2\left[\frac{p_{\min}}{1+p_{\min}}\right]^2\right) > 0. \tag{D3}$$

A yes instance reaches the ideal value with the perfect matching and uniform powers $\bar{p}$. A no instance remains at least δ_{box} below it. The reduction writes $KS = 2n^2$ constant-size channel indices in polynomial time; $P = K\bar{p}$ has polynomial encoding length. The fixed positive gaps are additive, not constant relative gaps. This continuous-power reduction establishes NP-hardness without an NP-membership claim for unrestricted continuous-threshold problems. □

S4.2 Proof of Theorem 2

Proof. Write $a = 1 - r$, $A = 1 + x$ and $B = 1 + y$. Differentiation gives

$$\begin{aligned} q_{xx} &= A^{-2} - \frac{2(1+ay)^2}{d^2}, \\ q_{yy} &= B^{-2} - \frac{2(1+ax)^2}{d^2}, \qquad q_{xy} = -\frac{2r}{d^2}. \end{aligned} \tag{D4}$$

At $x = 0$, the first expression tends to $1 - 2a^2$ as $y \to \infty$. If $a < 1/\sqrt{2}$, it becomes positive at a finite point, and by continuity at a nearby interior point. Thus global concavity is impossible.

For $a \ge 1/\sqrt{2}$, $A(1+ay) = d - ry$ and $ry/d \le r$ imply $A(1+ay)/d \ge a$; the two diagonal Hessian entries are nonpositive. The determinant numerator is

$$N = \left[2A^2(1+ay)^2 - d^2\right] \cdot \left[2B^2(1+ax)^2 - d^2\right] - 4r^2A^2B^2. \tag{D5}$$

Here $\det \nabla^2 q_r = N/(A^2B^2d^4)$, with strictly positive denominator. Section S2 lists every polynomial coefficient. They are nonnegative on $a \in [1/\sqrt{2}, 1]$, including the boundary, with strictly positive constant coefficient. Therefore $N > 0$, and both Hessian eigenvalues are negative at every finite point. □

S4.3 Proof of Theorem 3

Proof. Put $a = 1 - r$, $A = 1 + x$, and $k = 1 + ax$, and define

$$C = A(1 - \lambda_1 A), \qquad B = 2aA - k(1 + \lambda_1 A). \tag{D6}$$

The first interior stationarity equation, multiplied by the positive quantity Ad, is $C + By = 0$. If $C = 0$, then $\lambda_1 A = 1$ and $B = -2r \neq 0$; hence $B = C = 0$ is impossible. Every interior candidate therefore satisfies $y = -C/B$. Substitution into the second stationarity equation gives the polynomial

$$\Pi(x) = [2k - (1+\lambda_2)A]B^2 - [k - \lambda_2(A+k)]CB - \lambda_2 k C^2. \tag{D7}$$

Its degree is at most five, and the algebraic identity $\Pi(-1) = 2r^3 > 0$ shows that it is not identically zero. The value -1 is used only for this identity and is not feasible. Retain real roots with $L_x < x < U_x$, $B(x) \neq 0$, $L_y < -C(x)/B(x) < U_y$, and both original stationarity equations satisfied. Boundary roots are already covered by edges and corners. Multiplicity does not create additional candidate points.

On a horizontal edge $y = t$, set $h = 1 + at$ and $e = 1 + t$. The stationary equation is

$$2h(1+x) - (e + hx) - \lambda_1(1+x)(e+hx) = 0. \tag{D8}$$

This polynomial has degree at most two and cannot vanish identically: its quadratic coefficient is nonzero when $\lambda_1 \neq 0$, and its linear coefficient is $h > 0$ otherwise. Vertical edges are symmetric. Compactness and the first-order conditions on each face complete the enumeration. □

Degeneracies and numerical realization. The algebra applies to signed real prices, including zero; it also includes $r = 1$. At $r = 0$, $q_0 = \ln(1+x) + \ln(1+y)$ is separable, and the clipped scalar solutions in the main text apply. A zero-width interval leaves the other coordinate's edge equation; two zero-width intervals leave one point. In physical coordinates a zero gain eliminates the rate dependence on that coordinate, leaving only its linear price: choose the lower endpoint for a positive price, the upper endpoint for a negative price, and either for zero. For the nonnegative network multiplier, choosing minimum power is therefore valid even when both gains vanish.

For rational inputs, the algebraic oracle uses exact rational polynomials and SymPy real-root isolation. It repeatedly removes factors common to Π and B (none can represent a feasible interior stationary point), then takes the square-free part. Distinct repeated roots are retained once. Isolating intervals are intersected with the rectangle; $-C/B$ is enclosed only when the denominator interval excludes zero. Interval candidate boxes may conservatively include a nonstationary point, which cannot invalidate an upper bound. Feasible midpoints give lower bounds. Decimal divisions are rounded outward; logarithms, which use correctly rounded nearest evaluation, are widened with adjacent representable values. The implementation tries root widths $10^{-12}, 10^{-24}, 10^{-36}$ and returns only if the enclosing upper minus feasible lower is at most the requested positive tolerance. A denominator still containing zero or an unresolved gap raises an arithmetic error. The procedure therefore requires rational inputs and sufficient precision. The network experiments use the independently checked rectangular-partition algorithm in Section S5; the algebraic oracle is evaluated separately.

S4.4 Continuous configuration primal and weak duality

Let $\mathcal{R} = [p_{\min}, p_{\max}]^2$ and let μ_{uvs} be a finite nonnegative Borel measure on $\mathcal{R}$ for every $u < v, s$. Its coordinates represent p_u, p_v. The continuous configuration relaxation is

$$\begin{aligned} V_{\mathrm{CLP}} = \sup_{\mu \geq 0} \quad & \sum_{s,u<v} \int_{\mathcal{R}} q_{uvs}(p_u, p_v)\, d\mu_{uvs} \\ \text{s.t.} \quad & \sum_{s} \sum_{\{u,v\} \ni i} \mu_{uvs}(\mathcal{R}) = 1 \quad (i \in \mathcal{K}), \\ & \sum_{u<v} \mu_{uvs}(\mathcal{R}) = 1 \quad (s \in \mathcal{S}), \\ & \sum_{s,u<v} \int_{\mathcal{R}} (p_u + p_v)\, d\mu_{uvs} \leq P. \end{aligned} \tag{D9}$$

An integral feasible schedule places a unit Dirac mass at the selected powers for each selected pair/resource and zero measures elsewhere, proving $V_{\mathrm{C}} \leq V_{\mathrm{CLP}}$. The objective and moments are finite because the domain is compact and the rate continuous. Given unrestricted α_i, β_s and $\lambda \geq 0$ satisfying every pointwise inequality (8), integration and the mass constraints give $V_{\mathrm{CLP}} \leq \sum_i \alpha_i + \sum_s \beta_s + \lambda P$. Proposition 1 constructs such inequalities from complete pricing bounds. This uses weak duality only and makes no strong-duality assumption.

S4.5 Finite-tolerance best-bound refinement

Fix $\varepsilon > 0$, finitely many assignments, and exact arithmetic. A power box intersects the budget-feasible domain if and only if its lower corner satisfies the budget, since every coordinate is bounded below by that corner. Discard boxes failing this test. Before refinement, evaluate the unique feasible point of each singleton initial box and update the global incumbent L. Its exact singleton upper is then no larger than L. Whenever a feasible child is created, evaluate a feasible point in that child and update L. For such child evaluations, the implementation uses the budget-feasible center returned by its spatial-bound routine; a lower-corner evaluation is also admissible. Splits preserve coverage of the feasible domain; feasible points on a shared face may be included in both children without harm.

For either choice, the mixed-monotone bound and the rate at the selected feasible point differ uniformly by a quantity tending to zero with box diameter: both points lie in the same box and all rate expressions are uniformly continuous on the compact original box. The same holds for the affine tangent/chord bounds, since their derivatives are bounded there and their singleton limits agree with the rate; zero-width chords use point values. Taking their minimum with the mixed-monotone bound preserves this property. Thus a single $\delta > 0$ exists across the finite assignment family such that every evaluated feasible box of diameter below δ has upper bound at most its point value plus ε, hence at most $L + \varepsilon$ after the incumbent update.

The remaining initial boxes have positive diameter. Each is discarded by a valid bound or bisected when selected, so even initially small, unevaluated roots contribute only finitely many exceptions. Every feasible descendant is evaluated on creation, and the preceding small-diameter estimate applies to it. Longest-edge midpoint bisection permits only finitely many nodes with diameter at least δ: along any branch a selected coordinate is halved, and after a uniformly bounded depth every coordinate is shorter than $\delta/\sqrt{K}$. Binary branching and finitely many roots therefore give finitely many such nodes. Including the finite initial-root exceptions, best-bound selection can process only finitely many boxes with upper bound exceeding $L + \varepsilon$. It terminates when no such box remains. Improving L can only discard more boxes.

Singleton-root evaluation is an additional premise of this ideal procedure. The archived root initialization does not perform this incumbent update; the measured intervals $[0.05, 0.2]$ W have positive width. Finite work caps, incomplete assignment coverage, finite-precision arithmetic floors and wall-clock interruption can prevent tolerance attainment, while a valid fallback bound continues to cover the unresolved domain.

S5 Numerical methods and independent verification

This section describes the grid, spatial and configuration-dual reference implementations used in Section S6. The finite-box/integral method is specified in Section S11.

S5.1 Matching and power updates

The Matching–WMMSE comparator combines the receiver, weight and amplitude updates of I-WMMSE [5] with the matching step of uplink fractional programming [6]. We adapt these components to exactly-once service and the common power budget. Let $a = (s, t)$, $t \in \{1, 2\}$, index the $K = 2S$ slots, and let $\pi(a)$ denote the user assigned to a slot. For noise-whitened channel $\mathbf{h}_{ks}$ and amplitude $v_k = \sqrt{p_k}$, define

$$\mathbf{C}_s = \mathbf{I} + \sum_{a \text{ on } s} p_{\pi(a)} \mathbf{h}_{\pi(a)s} \mathbf{h}_{\pi(a)s}^H, \qquad \mathbf{u}_a = \mathbf{C}_s^{-1} \mathbf{h}_{\pi(a)s} v_{\pi(a)}. \tag{M1}$$

Hermitian linear solves compute the receivers. The scalar MSE and positive weight are

$$e_a = 1 - 2v_{\pi(a)} \Re\{\mathbf{u}_a^H \mathbf{h}_{\pi(a)s}\} + \mathbf{u}_a^H \mathbf{C}_s \mathbf{u}_a, \qquad w_a = e_a^{-1}, \tag{M2}$$

The sum rate equals the maximum of $\sum_a (\ln w_a - w_a e_a + 1)$ over receivers and positive weights. At fixed receivers and weights this expression is a lower bound touching the current rate.

At fixed user powers, the score for assigning user k to slot $a = (s, t)$ is

$$2w_a v_k \Re\{\mathbf{u}_a^H \mathbf{h}_{ks}\} - p_k \sum_{b \text{ on } s} w_b |\mathbf{u}_b^H \mathbf{h}_{ks}|^2. \tag{M3}$$

A perfect $K \times K$ assignment fills every slot and serves every user once, including when all scores are negative. Receivers and weights are then recomputed. For fixed assignment the amplitude update is

$$\begin{aligned} v_k(\lambda) &= \left[\frac{B_k}{A_k + \lambda} \right]_{\sqrt{p_{\min}}}^{\sqrt{p_{\max}}}, \\ A_k &= \sum_{a \text{ on } s(k)} w_a |\mathbf{u}_a^H \mathbf{h}_{ks(k)}|^2, \quad B_k = w_{a(k)} \Re\{\mathbf{u}_{a(k)}^H \mathbf{h}_{ks(k)}\}. \end{aligned} \tag{M4}$$

The bracket operator clips to the displayed amplitude interval. Zero power-update coefficients give the minimum amplitude. A common nonnegative multiplier is found with at most 80 bisections; the feasible high endpoint is retained when the sum-power constraint is active. Every measured rate decrease is rejected, including a decrease at roundoff scale. Candidate powers are converted to exact rational values and repaired to satisfy the budget before independent checking.

The deterministic start assigns users to sequential slots at power $\min(p_{\max}, P/K)$. Eight additional starts permute the users and draw independent residual powers uniformly in $[0, p_{\max} - p_{\min}]$. Residuals are scaled together when their sum exceeds $P - Kp_{\min}$, then $p_{\min}$ is added. Table S1 gives the stopping rules and subsequent exchanges used in the ray-traced study. These finite searches produce feasible local solutions. The full channel vectors are required because pair gains and correlations alone do not determine the receivers for candidate reassignments.

The source implementation and equation-to-code map are included with the reproduction material described in Section S10.

S5.2 Finite work and time limits

For the ray-traced reference experiments, the grid is $\mathcal{G} = \{0.05, 0.10, 0.20\}$ W. The grid stage has a 3-s limit, including separate native ILP and LP limits of 1.25 s. Pricing and spatial refinement each allow at most 1,200 splits. Full allocates at most one quarter of the remaining continuation time to pricing. Spatial search permits at most 3,000 assignment roots, sufficient to cover the $8!/2^4 = 2,520$ assignments at $K = 8$; any unvisited root retains the universal ceiling. The $K = 16$ Grid+pricing arm prices all pair rectangles without enumerating schedules. These limits bound computational effort and may stop refinement before the requested tolerance.

Table S1: Matching–WMMSE procedure for the ray-traced study.

1	Initialize sequential slots with uniform feasible power; generate eight additional starts using `SeedSequence([caller_seed,j])`, $j = 0, \ldots, 7$. Caller seeds are $1101 + i$ for O1 and $1201 + i$ for ASU case index i.
2	For each start, allow at most 128 iterations: compute receivers/weights by (M1)–(M2); maximize the complete user-slot scores (M3) by a perfect assignment; recompute receivers/weights; update all amplitudes by (M4). Negative assignment scores remain eligible.
3	Represent each binary64 power by its exact rational value, clip and repair total power by deterministic largest-slack reductions. Reject every rate decrease; stop when improvement is at most $10^{-10} + 10^{-9} \max(1, \lvert R_{\mathrm{old}} \rvert)$ or the search deadline is reached. Retain the best feasible start.
4	Perform at most two two-resource exchange sweeps. Enumerate the six allocations of four users into two labelled pairs, excluding the current allocation, with at most sixteen all-user power refinements per candidate; retain only measured improvements.
5	Stop search five seconds before the charged deadline. Independently check and save the retained candidate within the remaining allowance. Record the feasible rate and process completion separately from global-gap attainment.

S5.3 Spatial and configuration-dual procedure

The reference takes stored channels, a feasible power box/budget, a grid, a target ε, a deadline and work limits. It returns a feasible integral allocation, verified interval $[L, U]$ and termination status.

1. Verify a uniform sequential-pair allocation and the interference-free ceiling U_0 from the main article to initialize the bounds.

2. Solve the numerical grid IP/LP. Repair candidate powers, verify every incidence/power constraint and outward-enclose rates before updating L.

3. In Full, use at most one quarter of the continuation time for rectangular pricing. Cover every pair/resource box and apply Proposition 1 only after the full price list is verified.

4. Subject to the root limit, cover labelled assignments with spatial bounds, using eligible pair tangents and the budget-aware affine bound. Unvisited assignments retain fallback coverage.

5. While time and split limits allow, select a live box of largest bound. Discard infeasible or dominated boxes; otherwise bisect a longest power interval, evaluate each child's feasible center and retain covering child bounds. Verify improved candidates before updating L.

6. Reconstruct global coverage and intersect complete spatial/dual bounds with U_0. Return $[L, U]$ verified before the deadline. The target is attained only if $U - L \leq \varepsilon$ and the run returns normally within the watchdog limit; otherwise report the gap and termination reason.

S5.4 Construction and independent verification of bounds

The optimization program constructs candidate schedules and complete binary box partitions using high-precision logarithm approximations. A separate verifier checks the resulting bounds without importing that program or a numerical optimizer. It reconstructs the pair coefficients as exact rationals from stored binary64 channel entries and checks user coverage, resource occupancy, power bounds and total power. Logarithm enclosures use a 100-bit integer fixed-point atanh expansion with forty terms and an explicit positive tail bound. Every partition split is checked, every terminal region contributes a valid bound, and unlisted spatial roots retain a universal bound. The repaired dual can improve the initial universal bound only after every pair/resource rectangle has been priced.

Each completed check produces a verification record containing the case, candidate and proof hashes and the time at which the result was saved. Files receive unique names after writing is complete. Only bounds checked and saved by the charged deadline can improve the reported interval; timely bounds for the same case may be intersected. Numerical failures remain recorded even when an earlier valid bound is available. The primary endpoint also requires normal process return by the external deadline.

Saved-data verification checks every case, candidate, execution and verification-record hash together with the timing records. A previously reconstructed result is reused only for an identical case and identical canonical schedule/proof. Timed experiments clear the verifier cache before each method execution. The resulting bounds apply to the stored nominal coefficients, including their quantization; channel-estimation uncertainty is outside this model.

S5.5 Exact serialization, arithmetic and rejection rules

Case binding uses the SHA-256 of UTF-8 JSON with sorted keys and compact comma/colon separators. Powers, gains and certified bounds are serialized as rational strings; stored binary64 real/imaginary channel values are reconstructed exactly with integer ratios. The checker does not certify the pre-quantization physical field. It derives the complete partitions itself from rational split events and rejects duplicate/missing rows, illegal splits, inconsistent channels, infeasibility and unsupported fields. Missing spatial roots require explicit universal fallback coverage; incomplete pair pricing cannot produce a repaired-dual bound.

For a positive rational $z = 2^k q$ with $1 \le q < 2$, put $t = (q-1)/(q+1) \in [0, 1/3)$. The independent checker uses

$$\ln q = 2\sum_{j=0}^{39} \frac{t^{2j+1}}{2j+1} + \mathcal{R}, \quad 0 \le \mathcal{R} \le \frac{2(1/3)^{81}}{81(1-1/9)}. \tag{M5}$$

Each operation is rounded outward by integer floor/ceiling on the 2^{-100} grid; $\ln 2$ is enclosed similarly. Negative range exponents interchange endpoints. An invalid certificate or an incomplete check leaves the earlier verified fallback as the usable bound. Bound validity, completion before the deadline, target attainment and normal process return are recorded separately. The algebraic oracle uses the rational-root and directed-Decimal procedure in Section S4; the network checker verifies complete box partitions.

S5.6 Notation

Table S2: Notation used in the article and supplementary methods.

Symbol	Meaning and domain
$K = 2S, M$	User count, resource count and receive dimension.
$\mathcal{K}, \mathcal{S}, \mathcal{E}$	Users, orthogonal resources and unordered user pairs.
$a_i, \mathbf{h}_{is}, \mathbf{n}_s$	Unit-variance transmitted symbol, whitened channel and unit-covariance complex Gaussian noise.
$p_i, p_{\min}, p_{\max}, P$	Transmit powers, individual limits and aggregate radiated-power budget, in W.
g_{is}, r_{uvs}, x, y	Squared channel gain, squared normalized correlation and normalized received-power coordinates.
q_r, q_{uvs}, e_s	Generic pair log-rate, physical-power pair log-rate, and pair selected on resource s.
$Q, Q/S$	Sum of resource SEs in bit/channel-use and gross occupied-band SE in bit/s/Hz.
B_{r}, η	Resource bandwidth in Hz and useful-data fraction.
$z, c, \mathcal{G}$	Binary assignment, pair/resource/power configuration and finite power grid.
$\alpha_i, \beta_s, \lambda$	Unrestricted user/resource prices and nonnegative shared-power price.
$V_{\mathrm{C}}, V_{\mathrm{CLP}}$	Continuous integral-schedule optimum and measure-LP relaxation value, in nat.
L, U	Verified feasible lower value and retained global upper value, in nat.
$U_{\mathrm{dual}}, U_{\mathrm{MM}}, U_0$	Repaired dual bound, mixed-monotone box bound and interference-free fallback, in nat.
$B(x), C(x), \Pi(x)$	Pricing-elimination polynomials; Π is distinct from power budget P.
L_j, U_j, λ_j	Pricing-coordinate interval and price, $j \in \{x, y\}$.

Symbol	Meaning and domain
R_i, J	User contribution to occupied-band SE in bit/s/Hz and dimensionless Jain index.

S6 Ray-traced evaluation: setup and results

We evaluate the scheduling methods on the O1_60 and ASU_Campus1 scenes generated with DeepMIMO [7]. Each scene uses one access point with eight receive antennas. The physical settings in Table S3 correct a delay-window mismatch identified in an earlier study. The original and corrected samples have separate prespecified analyses; their results are reported below without pooling.

S6.1 Channel generation and physical normalization

Table S3: Physical settings of the corrected one-AP study. Array shapes follow DeepMIMO's three-coordinate convention.

Parameter	Setting
Scene / carrier	O1_60 / 60 GHz; ASU_Campus1 / 3.5 GHz
Active access point	AP 1 in each scene
AP / user array	$[1, 8, 1]$ / $[1, 1, 1]$; isotropic elements
AP spacing / rotation	0.5 wavelength / $[0, 0, 0]$ degrees
AP field of view	$[360, 180]$ degrees
Paths / additional effects	Up to ten paths; Doppler and dual polarization disabled
Generator user rows	O1: 321–340; ASU: 21–40
Transform / stored tones	512 over 50 MHz / $0, 16, \ldots, 496$
Main / scale FFT indices	$0, 128, 256, 384$ / $0, 64, \ldots, 448$
Tone spacing / assumed CP	97.65625 kHz / 512 samples (10.24 μs)
Noise / noise figure	-174 dBm/Hz / 7 dB; 1.94850×10^{-15} W per tone
Power box / total budget	$[0.05, 0.2]$ W / $K/10$ W
Cases per scene	Twelve with $(K, S) = (8, 4)$; four with $(16, 8)$

The corrected model uses 512 FFT subcarriers over 50 MHz, with 32 stored tones indexed $0, 16, \ldots, 496$. The useful symbol and assumed 512-sample cyclic guard are each 10.24 μs. We assume ideal synchronization, zero carrier-frequency offset, block-static channels and no inter-resource interference. Every retained absolute delay must be nonnegative and below the common guard/window, with no per-user delay shift. An input or guard violation invalidates the complete scene rather than triggering resampling. The settings were specified before observing the corrected outcomes. A time-domain finite-guard receiver is not simulated.

The array tuple follows X, Y, Z order, so $[1, 8, 1]$ places eight half-wavelength-spaced elements along Y. In the generator's `OFDM_PathGenerator.no_LPF` routine, a path with power P_ℓ, phase ϕ_ℓ and normalized delay n_ℓ contributes

$$\sqrt{P_\ell/N}\exp\{\mathrm{j}(\phi_\ell - 2\pi n_\ell k/N)\}, \qquad N = 512. \tag{E1}$$

Multiplication by $\sqrt{N}$ reverses this path-amplitude normalization. Generator complex64 outputs are promoted to complex128 and multiplied by $\sqrt{512}/\sqrt{\sigma^2}$, where $\sigma^2 = 1.9484983544617996 \times 10^{-15}$ W per physical tone follows from -174 dBm/Hz and a 7-dB noise figure. Verification uses the resulting stored binary64 arrays, including the earlier generator quantization.

The physical tone spacing is $50\,\mathrm{MHz}/512 = 97.65625$ kHz. Main resource positions $0, 8, 16, 24$ in the stored array correspond to FFT indices $0, 128, 256, 384$; scale positions $0, 4, \ldots, 28$ correspond to $0, 64, \ldots, 448$. The 32 stored tones span an aggregate bandwidth of 3.125 MHz, while the four and eight selected tones occupy 0.390625 and 0.78125 MHz, respectively. These tones are spaced across the FFT grid. We report gross occupied-tone SE, $Q/S = L/(S \ln 2)$, excluding the guard and other overheads. The equal-length guard alone would halve payload SE. The waveform is an ideal OFDM benchmark rather than a standardized framing, and its normalization must be held fixed in method comparisons.

S6.2 Sampling and computational allowances

The corrected pools use generator rows 321–340 in O1 and 21–40 in ASU. Their coordinates are required to be disjoint from the original study and earlier O1 development and CSI pools. Each network draws distinct users, with replacement across networks conditional on a fixed scene. Zero and weak users are retained. Each scene contributes twelve $K = 8, S = 4$ networks and four $K = 16, S = 8$ networks. Main user seeds are 202611310/311 and scale seeds are 202611320/321. Method order uses seed 202611330; the eight repeated analytic execution checks use order seed 202611350. The eight original synthetic mechanism controls are reported once as separate diagnostic evidence.

Main executions receive an 18-s charged allowance and a 20-s external limit; scale executions receive 28/30 s, and analytic checks receive 8/10 s. Algorithms, grid levels, power domains, receiver assumptions and split/root limits remain fixed between the original and corrected studies. A Python 3.12 controller enforces the original 3,600-s overall limit; the Python 3.9 generator receives relative watchdog durations. Complete method blocks are allocated before execution, and incomplete blocks are recorded as missing. No case is replaced after observing its outcome.

S6.3 Endpoints and statistical analysis

The primary comparison is the equal-scene difference between Full and Curvature in the probability of a verified gap at most 0.01 nat saved by the charged deadline, with normal process return by the external limit. Let $Y_{sj}^F, Y_{sj}^C \in \{0, 1\}$ denote these outcomes for paired case j in scene s. The estimand and its estimate are

$$\Delta = \frac{1}{2}\sum_s \{\Pr(Y_s^F = 1) - \Pr(Y_s^C = 1)\}, \qquad \hat{\Delta} = \frac{1}{2}\sum_s \frac{n_{10,s} - n_{01,s}}{n_s}, \quad n_s = 12. \tag{E2}$$

Here $n_{10,s}$ counts success by Full alone, and $n_{01,s}$ counts success by Curvature alone. For each of the four discordant probabilities, the Clopper–Pearson limits [8] for count k out of n are

$$\ell(k,n) = \begin{cases} 0 & k = 0, \\ F^{-1}_{\mathrm{Beta}(k,n-k+1)}(0.00625) & k > 0, \end{cases} \quad u(k,n) = \begin{cases} 1 & k = n, \\ F^{-1}_{\mathrm{Beta}(k+1,n-k)}(0.99375) & k < n. \end{cases} \tag{E3}$$

Each interval has confidence at least 98.75%. A Bonferroni bound over the four intervals gives at least 95% simultaneous coverage [9], without requiring independence among them. The scene interval is $[\ell_{10,s} - u_{01,s}, u_{10,s} - \ell_{01,s}]$; averaging its endpoints across the two scenes gives the interval for Δ. A missing primary paired outcome makes its scene estimate undefined and its interval $[-1, 1]$. All 24 primary pairs are observed. With zero discordances, the overall endpoints are $\pm[1 - 0.00625^{1/12}]$, or ± 0.3449 after rounding.

Secondary scalar differences are resampled within each scene and averaged with equal scene weights. We use the empirical 0.025 and 0.975 quantiles of 10,000 replicate means [10]. The random stream is `default_rng(202611231)`, used first for SE Full–Matching–WMMSE and then for gap Full–Curvature, with O1 preceding ASU for each contrast. Only complete valid scalar pairs enter these exploratory intervals; missing values are not set to zero. Rate, gap, fairness, power, time and memory summaries are descriptive. Sample size was limited by the computational budget rather than a prospective power calculation. Inference is conditional on the two scene pools, and an interval containing zero does not establish equivalence.

Saved-data analysis. All 128 corrected-study executions finished before postprocessing. The automatic analysis exceeded its 5-s limit, and the combined analysis and proof-verification stage remained incomplete after 59.302 s of its 60-s allowance. A separately allotted 120-s saved-data calculation completed in 92.894 s. This increased the original aggregate analysis allowance of 120 s while remaining within the 3,600-s overall limit. The calculation used unchanged analysis and verification code on hash-checked data; no channel, optimizer output, sample or endpoint was changed. The original incomplete analyses remain in the archive.

S6.4 Original study and the delay-window error

The original waveform used 32 tones over 50 MHz, giving a $0.64\,\mu$s useful-symbol window. In the selected ASU rows, 5,694 users had 53,786 retained paths with delays between 0.817592 and 2.60087 μs; another

2,526 users had no retained paths. The generator clipped every retained path beyond the window, leaving all 8,220 pool channels equal to zero. This input/model mismatch motivated the longer window specified above. The zero channels do not indicate physical lack of coverage, and the original observations remain in their prespecified analysis.

All 128 original executions returned normally, with 590 timely verification records reconstructed independently. Full and Curvature both missed the target on all twelve O1 cases and attained it trivially on all twelve all-zero ASU cases. The equal-scene primary difference was zero with conservative 95% interval $[-0.3449, 0.3449]$. The ASU outcomes therefore provide no evidence of certification on nondegenerate channels.

In the original O1 study, Full and Curvature have mean gaps of 0.569520 and 0.572798 nat, respectively; Full and Grid have identical main rates. The four O1 scale cases complete all 960 prices and reduce the mean Grid gap from 40.43495 to 7.06737 nat while leaving achieved rates unchanged.

S6.5 Corrected study: target attainment and achieved rates

All 96 main, 24 scale and eight analytic reliability executions returned normally, with no recorded solver, storage or verification-timeout error. No main or scale arm attained the 0.01-nat target. Curvature and Full each certified both repetitions of the analytic fixture; Grid and Matching–WMMSE did not. These repeated checks assess implementation reliability rather than independent propagation samples. In each scene, the primary paired counts are $n_{00} = 12, n_{01} = n_{10} = n_{11} = 0$, giving a difference of zero and a 95% interval of $[-0.3449, 0.3449]$ for this nondegenerate population.

Table S4 gives descriptive means for each scene. The complete 128-row export contains every execution, its bounds, duration, power, fairness, completion status and proof coverage. Achieved SE is $L/(S \ln 2)$, while the absolute gap $U - L$ is in nat. Numerical ILP/LP values and statuses are retained alongside the independent feasibility and continuous-bound checks.

Table S4: Corrected-study means; $n = 12$ per main arm and scene, $n = 4$ per scale arm and scene.

Panel	Scene	Arm	SE	Gap (nat)	Time (s)	Jain
Main	O1	Curvature	27.33527	0.61657	3.680	0.9947
Main	O1	Full	27.33527	0.59237	5.198	0.9947
Main	O1	Grid	27.33527	19.78376	0.584	0.9947
Main	O1	Matching–WMMSE	27.18806	20.19191	2.605	0.9940
Main	ASU	Curvature	11.44830	0.63401	3.814	0.5840
Main	ASU	Full	11.44830	0.54735	5.340	0.5840
Main	ASU	Grid	11.42849	20.96990	0.581	0.5694
Main	ASU	Matching–WMMSE	11.34300	21.20696	2.928	0.5930
Scale	O1	Grid	27.82688	38.86266	0.999	0.9951
Scale	O1	Matching–WMMSE	27.49514	40.70220	5.907	0.9941
Scale	O1	Grid+pricing	27.82688	7.22491	3.412	0.9951
Scale	ASU	Grid	9.59812	38.58020	0.930	0.5957
Scale	ASU	Matching–WMMSE	9.42257	39.55367	8.634	0.6185
Scale	ASU	Grid+pricing	9.59812	3.93320	3.290	0.5957

All main solutions use 0.8 W and all scale solutions use 1.6 W. Full and Curvature rates coincide in all 24 main cases. Shared refinement improves on Grid in ten ASU cases and ties in two, while all twelve O1 cases tie. Full has tighter upper bounds than Curvature in two O1 and nine ASU cases; the other thirteen tie. Its main gaps range from 0.404865 to 0.736759 nat in O1 and from 0.325381 to 0.818872 nat in ASU.

Scale pricing covers 960/960 rectangles in all eight cases, but gaps remain 7.150558–7.444041 nat in O1 and 3.373034–4.914289 nat in ASU.

The equal-scene Full-minus-Matching–WMMSE mean is 0.126257 bit/s/Hz, with exploratory interval $[0.0766, 0.1768]$. Grid contributes 0.116355 bit/s/Hz (92.2%) and shared refinement contributes 0.009902 bit/s/Hz. The equal-scene Full-minus-Curvature gap difference is -0.0554 nat, with paired-bootstrap interval $[-0.0851, -0.0295]$. Across the same 24 Full outcomes, the descriptive relative gaps $100(U - L)/L$ range from 0.52% to 2.74%. For $L > 0$, $L \le V_{\mathrm{C}} \le U$ implies $(V_{\mathrm{C}} - L)/V_{\mathrm{C}} \le (U - L)/L$, so these percentages conservatively bound unknown regret. At sixteen users, mean relative gaps decrease from 25.20% to 4.68% in O1 and from 74.58% to 7.47% in ASU when pricing replaces the interference-free ceiling; all absolute gaps remain above 0.01 nat.

S6.6 Where the curvature condition applies

For each saved case, we counted all candidate pair/resource combinations satisfying $r \le 1 - 1/\sqrt{2}$, using the exact equivalent test $2(1 - r)^2 \ge 1$. Averaging the case fractions gives 1.12% (O1) and 1.79% (ASU) for the main cases, and 0.94% and 11.64%, respectively, for the scale cases. The denominators include all candidates, not only selected pairs. These descriptive summaries use the existing case records and do not constitute new channel trials or explain algorithmic performance by themselves.

S6.7 Controlled mechanism and isolated oracle diagnostics

The eight synthetic cases in Fig. S4 have $K = 4, S = 2, M = 4$ and channels

$$\mathbf{h}_{us} = \sqrt{G/4}\big(\sqrt{c}\,\mathbf{z}_{cs} + \sqrt{1 - c}\,\mathbf{z}_{us}\big).$$

All underlying vectors are independent $\mathcal{CN}(\mathbf{0}, \mathbf{I}_4)$ draws, except that $\mathbf{z}_{cs}$ is shared by the users on resource s. Resources are independent. The mixture parameter is $c \in \{0, 0.85\}$, the gain scale is $G \in \{1, 100\}$, and seeds are 202611240/241. The parameter c differs from the realized squared correlation r; $c = 0$ gives independent channels rather than exact orthogonality. Table S5 lists the cases and verified continuation gaps in plot order. Each arm starts from the same grid incumbent and numerical dual. A 2-s search is followed by independent checking, so the total diagnostic duration may exceed 2 s. These controls isolate bound behavior under common initialization; their timing is separate from the charged network comparison.

Table S5: Original synthetic controls, reused once. Gaps are in nat; no display floor is applied to this table.

ID	c	G	Seed	Reference	Curvature	Full
1	0.0	1.0	202611240	5.25e-28	5.20e-28	5.17e-28
2	0.0	1.0	202611241	1.42e-09	5.41e-15	8.66e-14
3	0.0	100.0	202611240	2.57e-04	3.45e-05	3.45e-05
4	0.0	100.0	202611241	1.43e-04	1.04e-09	1.04e-09
5	0.85	1.0	202611240	5.15e-28	5.15e-28	5.15e-28
6	0.85	1.0	202611241	5.10e-28	5.10e-28	5.10e-28
7	0.85	100.0	202611240	1.67e-06	1.67e-06	1.67e-06
8	0.85	100.0	202611241	2.69e-03	2.69e-03	2.69e-03

All eight diagnostic continuations meet the target. Eligible tangents tighten selected low-alignment, high-gain cases; Full's repaired dual never improves on its final spatial bound in these eight cases. The isolated oracle uses the first pair on the first resource of each synthetic case, its grid-dual price and the physical power box $[0.05, 0.2]^2$ W. All eight calls return within their 2-s watchdog with verified enclosures narrower than 10^{-7} nat. Oracle computation and checked generic pricing use different timing boundaries, so these measurements do not determine a matched speedup or network-level benefit.

S6.8 Solved finite-grid ILP and LP instances

Table S6 gives both numerical master values for every corrected-study network. Status 0 denotes the numerical solver's reported optimum; continuous certification still requires the independent bounds. Each solve has a 1.25-s native limit within the shared method allowance. Main masters contain 1,008 configurations and scale masters 8,640. Distinct ILP and LP values reveal a finite-grid relaxation gap even when both solves report success.

Table S6: Corrected-study grid ILP and LP objective values in nat, for all 32 networks.

Case	ILP	LP	ILP status	LP status
asu-main-00	32.208132	32.263654	0	0
asu-main-01	25.040614	25.213387	0	0
asu-main-02	33.415336	33.768411	0	0
asu-main-03	21.062004	21.062004	0	0
asu-main-04	44.934721	44.934721	0	0
asu-main-05	34.580210	34.585543	0	0
asu-main-06	25.822444	25.850471	0	0
asu-main-07	29.243989	29.246271	0	0
asu-main-08	20.467787	20.467787	0	0
asu-main-09	45.079992	45.080084	0	0
asu-main-10	27.334578	27.386179	0	0
asu-main-11	41.048373	41.048373	0	0
asu-scale-00	46.257977	46.257977	0	0
asu-scale-01	72.381528	72.450897	0	0
asu-scale-02	45.699366	45.801507	0	0
asu-scale-03	48.554231	48.554231	0	0
o1-main-00	77.606612	77.636891	0	0
o1-main-01	75.003805	75.039337	0	0
o1-main-02	76.204076	76.204076	0	0
o1-main-03	74.778471	74.778471	0	0
o1-main-04	76.718514	76.718514	0	0
o1-main-05	74.040659	74.040659	0	0
o1-main-06	76.761765	76.761765	0	0
o1-main-07	73.306619	73.306619	0	0
o1-main-08	77.646626	77.646626	0	0
o1-main-09	77.386773	77.386773	0	0
o1-main-10	75.594369	75.594369	0	0
o1-main-11	74.425319	74.425319	0	0
o1-scale-00	155.844316	155.844316	0	0
o1-scale-01	152.390440	152.684282	0	0
o1-scale-02	155.849359	155.849359	0	0
o1-scale-03	153.135755	153.135755	0	0

S6.9 Minimum-user rates and fairness

Table S7 reports descriptive means across all twelve main cases per scene. Full and Curvature increase ASU sum SE relative to Grid, but their mean minimum-user SE is lower: 0.1530 versus 0.1602 bit/s/Hz. Matching–WMMSE gives 0.1958 bit/s/Hz with a lower sum SE. Thus sum-rate improvement does not imply improved weakest-user service. Jain's index [11] is undefined for an all-zero rate vector, while its minimum-user rate is zero; none of the listed corrected-study indices is missing. No QoS or fairness constraint is imposed.

S6.10 Case-level distributions and rate decomposition

The following figures retain every main and scale case, together with the separate synthetic controls. Grid accounts for 92.157% of the equal-scene rate gain from Matching–WMMSE to Full; the shared refinement

Table S7: Mean minimum-user occupied-band SE and Jain index, twelve cases per row.

Scenario	Method	Min. SE (bit/s/Hz)	Jain index
O1	Grid	3.0433	0.9947
O1	Curvature	3.0433	0.9947
O1	Full	3.0433	0.9947
O1	Matching–WMMSE	3.0155	0.9940
ASU	Grid	0.1602	0.5694
ASU	Curvature	0.1530	0.5840
ASU	Full	0.1530	0.5840
ASU	Matching–WMMSE	0.1958	0.5930

contributes 0.009902 bit/s/Hz overall. Tighter upper bounds and higher achieved rates are distinct effects.

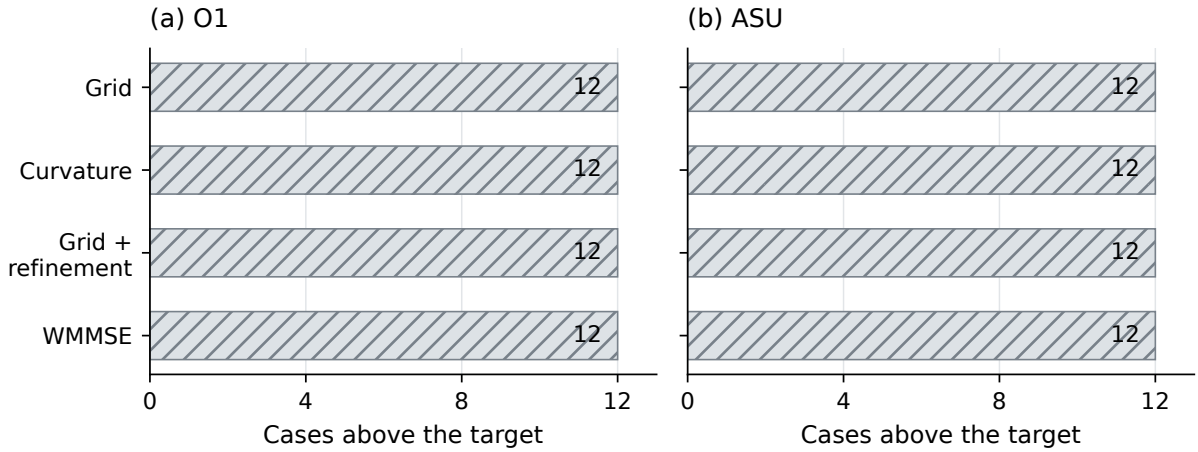


Figure S1: Target outcomes for all twelve corrected eight-user cases per scene and method under the 18-s allowance. "Grid + refinement" denotes Full, and "WMMSE" denotes Matching–WMMSE. Every case misses the 0.01-nat target; abnormal returns and missing outcomes are zero.

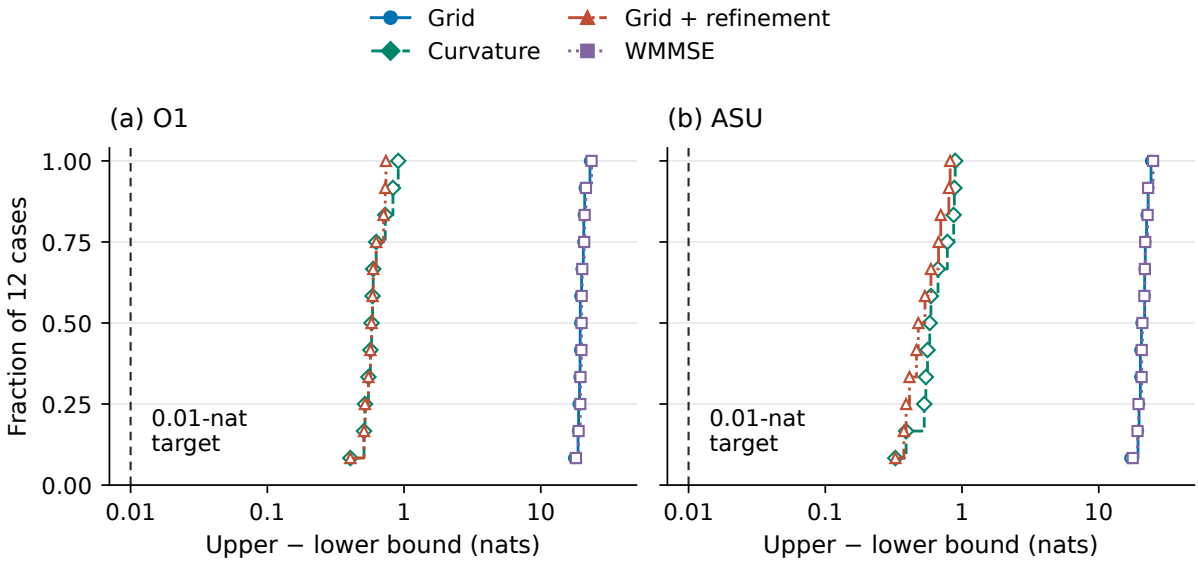


Figure S2: Empirical distributions of absolute upper-minus-lower gaps, with all twelve cases per scene and method. The dashed line marks the 0.01-nat target. Curvature and Grid + refinement identify the corresponding pipeline components; a leftward shift indicates tighter bounds.

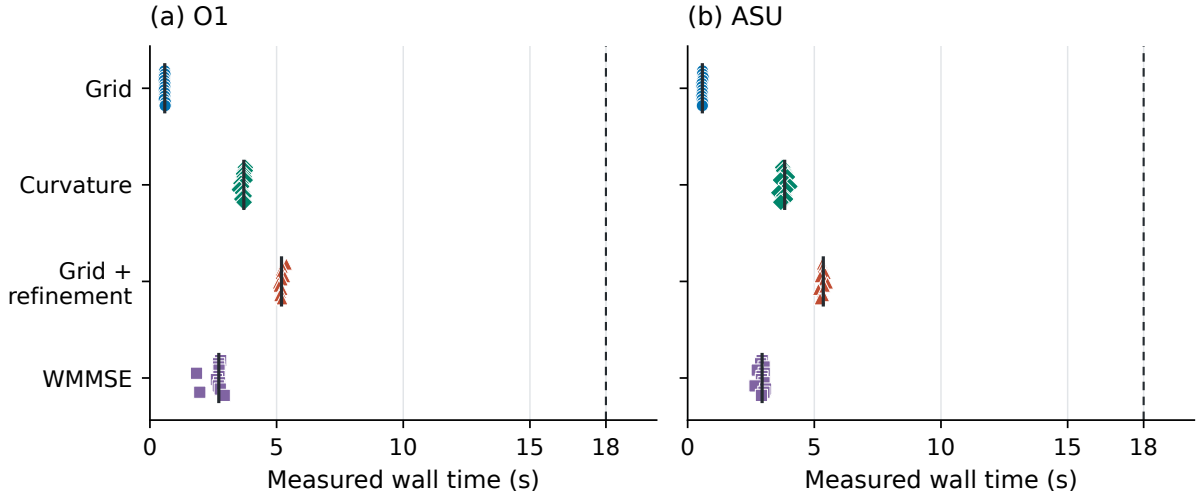


Figure S3: Main-case elapsed times, including initialization, search, independent checking and saving results. Each method has twelve observations per scene. Black marks show medians, and the dashed line is the 18-s charged limit. These capped durations were measured on one nondedicated host.

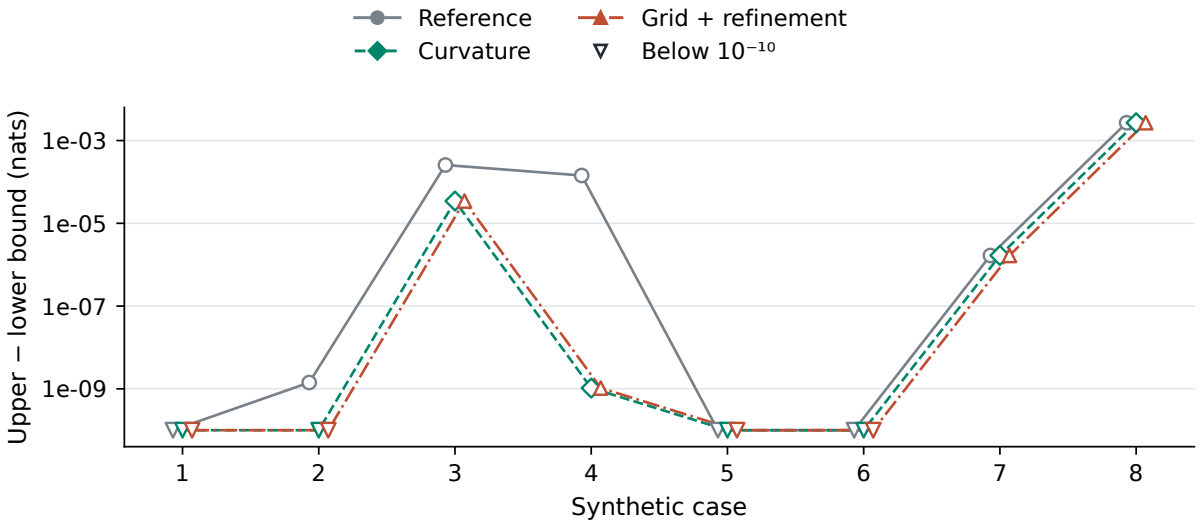


Figure S4: Eight synthetic mechanism controls with a common incumbent and 2-s searches followed by verification. The logarithmic display floor is 10^{-10} nat; downward open triangles mark values below that floor. These cases are separate from the ray-traced population.

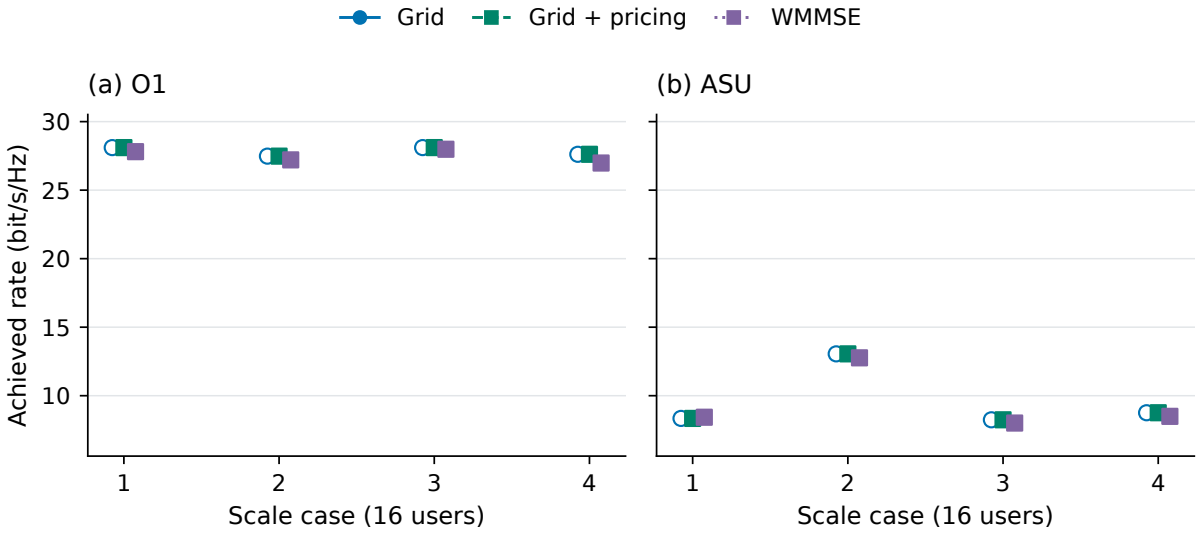


Figure S5: Gross occupied-tone SE for all eight sixteen-user cases under the 28-s allowance. Grid and Grid + pricing give equal rates; open Grid symbols and horizontal offsets distinguish coincident observations without changing their vertical positions. Case IDs identify fixed inputs.

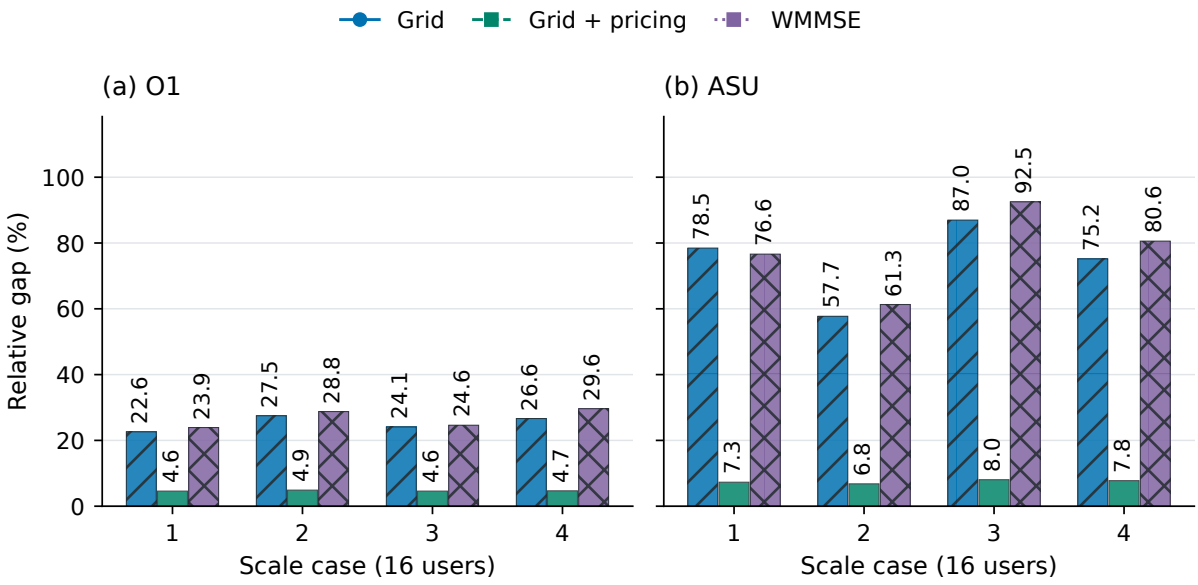


Figure S6: Relative global gaps, $100(U - L)/L$, for all eight sixteen-user cases and three methods. These percentages bound unknown regret; bar labels are rounded to one decimal place. All pricing runs cover 960/960 rectangles, and none attains the absolute target.

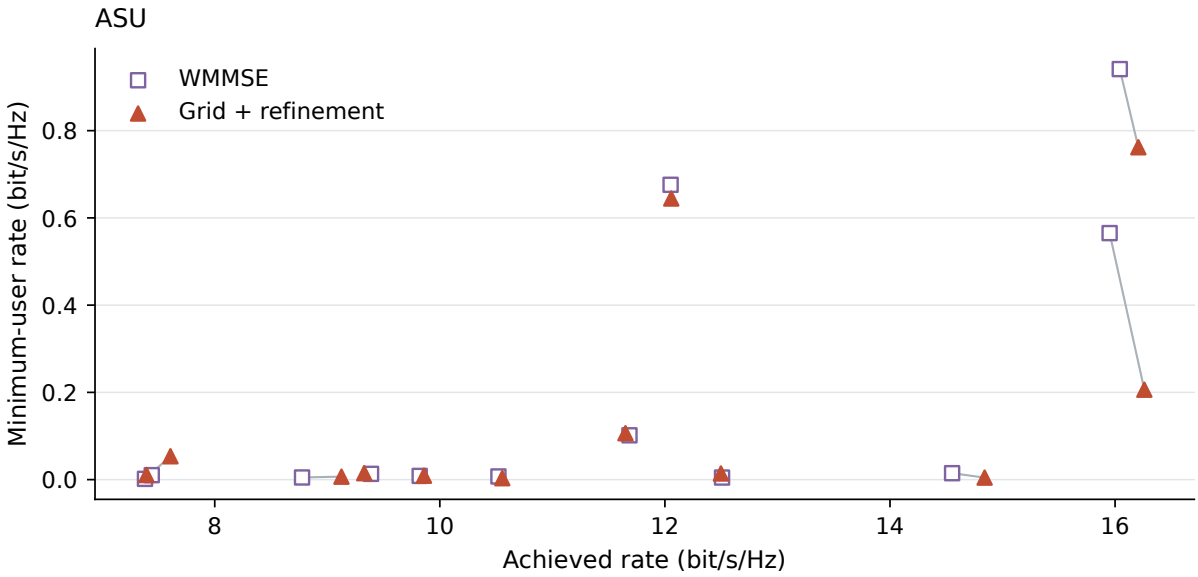


Figure S7: Achieved and minimum-user rates for all twelve ASU main cases. Each segment joins the WMMSE and Grid + refinement allocations of one case; higher achieved sum SE can accompany lower minimum-user SE.

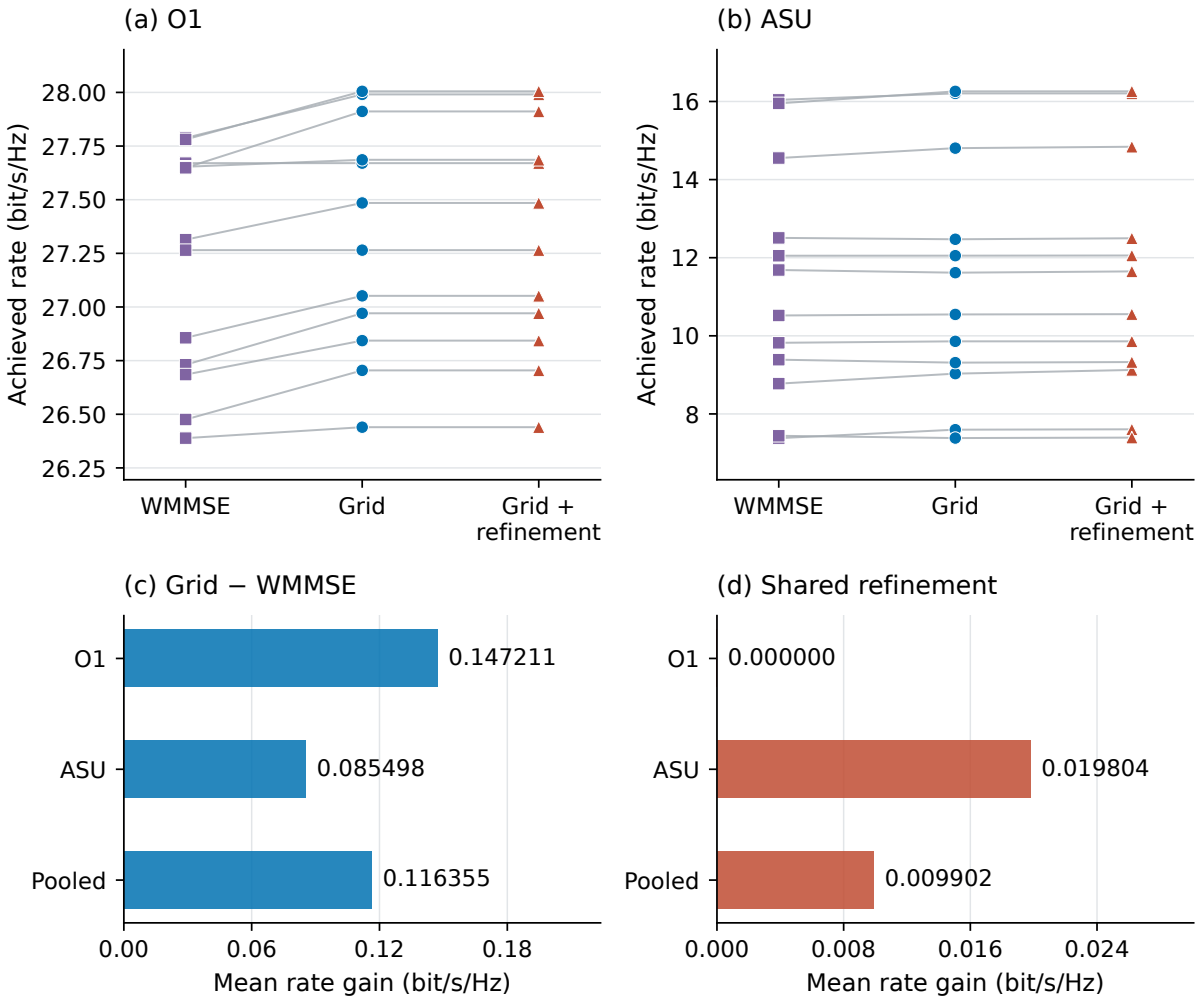


Figure S8: Paired achieved rates for all 24 main cases and their mean decomposition. Each gray line joins WMMSE, Grid and Grid + refinement for one case. Grid supplies 92.157% of the equal-scene gain, while shared refinement contributes 0.009902 bit/s/Hz overall.

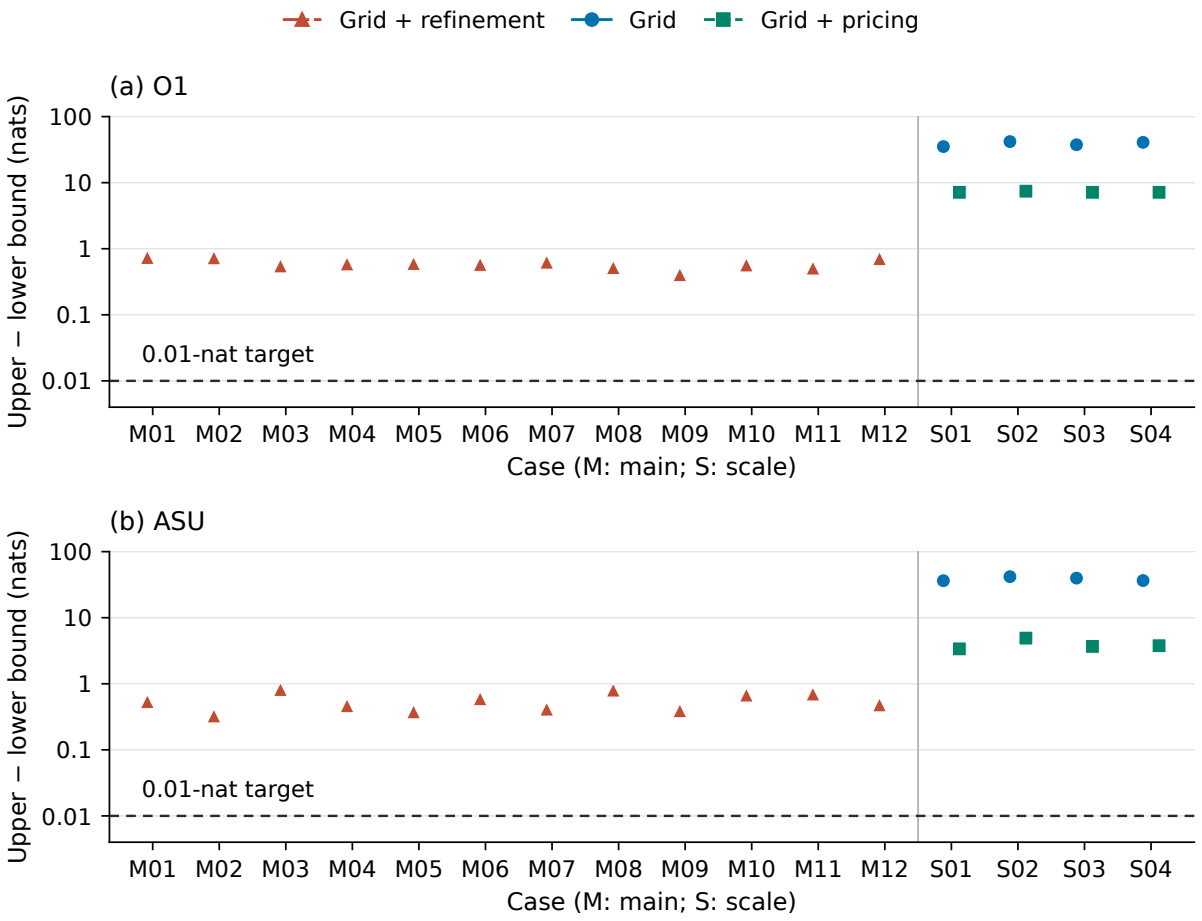


Figure S9: Absolute upper-minus-lower gaps for all 24 main and eight scale cases, separated within each scene. Main cases show Grid + refinement; scale cases show Grid and Grid + pricing. Every displayed interval remains above the dashed 0.01-nat target.

S7 Common-model power-component methods

We adapt power-search components from Fu et al. [12] and Thu et al. [13] to the instantaneous no-SIC MMSE rate, exactly-once service and common power constraints. These adaptations allow comparison under the same objective and feasible set. They retain the specified power updates rather than the sources' full architectures or guarantees. The controlled-array study combines them with common multistart and assignment-exchange procedures and compares them with power transfer, Matching–WMMSE and analytic-gradient SLSQP.

S7.1 Successive convex approximation

Fu et al.'s Eq. (25c) supplies a quadratic-over-linear minorant. For fixed unit-norm MMSE directions, phase-align the desired coefficients $b_i \geq 0$, set $v_i = \sqrt{p_i}$ and let $N_i(v) = 1 + \sum_{j \neq i} c_{ij} v_j^2$, where c_{ij} is the fixed-filter interference gain on the same resource. At the current point $\bar{v}$, put $a_i = b_i \bar{v}_i / N_i(\bar{v})$ and

$$t_i(v) = 2a_i b_i v_i - a_i^2 N_i(v).$$

The exact identity

$$\frac{b_i^2 v_i^2}{N_i(v)} - t_i(v) = \frac{[b_i v_i - a_i N_i(v)]^2}{N_i(v)} \geq 0$$

shows equality and gradient matching at $\bar{v}$. Maximizing $\sum_i \ln(1 + t_i)$ over the amplitude bounds, $\sum_i v_i^2 \leq P$ and $t_i \geq 0$ is a convex optimization problem. The last constraint restricts the surrogate to a conservative inner domain. MMSE directions are recomputed between power blocks and when evaluating the rate. Finite numerical and time limits can stop the updates before convergence.

The printed Eq. (25e) adds the previous denominator to interference plus noise, although Algorithm 2 initializes that quantity as the full denominator. To preserve the tangent identity above, we use the exact common-model denominator $N_i(v)$. The source's deterministic-equivalent analysis, LP-RZF regularization, QoS requirements and per-EDU constraints belong to its statistical-CSI downlink model and are not part of this adaptation.

S7.2 Improved differential evolution

Thu et al.'s Section IV-B, Eqs. (31)–(36) and Algorithm 1 provide two differential-evolution mutations, adaptive operator probabilities, forced-coordinate binomial crossover, an archive and gain-weighted Lehmer updates of mutation and crossover parameters. These locators refer to the accepted manuscript identified in the source metadata. We encode normalized scalar powers and project candidates onto the common power box and sum budget. Selection uses the recomputed-MMSE rate. The released method specification gives population and archive sizes, the top-candidate pool and choices left unspecified by the source. Effort counts include every operator invocation, following the source prose; Algorithm 1 instead places the count in the success branch. The source's global-probability argument is not transferred because its uniform offspring-density assumption does not follow from these differential-evolution operations.

S7.3 Relationship to the other comparators

Ou et al.'s [14] transferable power block reduces to the WMMSE amplitude quadratic in the Shannon limit after receiver conventions are aligned; counting it as a separate modern arm would duplicate the same update while removing its finite-blocklength contribution. Chen et al.'s [15] grouping and hybrid-beamforming design remains architectural context because stripping its beam-squint/RF variables would discard its principal coupled mechanism.

In the joint search of Section S8, T, SLSQP, Fu-SCA and Thu-IDE have the same permitted starts and four-user exchanges. Each power call receives the remaining search time, so a costly initial call can leave little time for assignment changes. Matching–WMMSE additionally retains its matching/receiver steps and seeded starts. All methods include initialization in their allowance and use the same independently checked anchor. Section S8.6 reports completed search work; Section S9 compares the power components on fixed assignments using the same 44 cases. The latter reuses the controlled-array sample rather than supplying another independent holdout.

S8 Controlled-array joint scheduling and power allocation

We use controlled arrays to compare feasible rate improvements across channel-gain and angular-spread regimes. The hypotheses, seed groups, methods, endpoints and computational allowances were specified before input generation. The 44 evaluation cases comprise eleven independent seed groups and are separate from the ray-traced study. The fixed-assignment comparison in Section S9 reuses these same cases and groups.

S8.1 Population and physical coefficients

There are $K = 8$ users, $S = 4$ labelled resources and $M = 8$ receive coordinates, with $p_i \in [1/20, 1/5]$ W and $P = 4/5$ W. Each seed group supplies four regimes: nominal SNR of 0 or 30 dB at 0.1 W, crossed with angular half-widths of 3° or 60°. The pilot seed, 2026113301, is separate from evaluation seeds 2026113401–2026113411, which were held out during method development. Four earlier ray-traced cases, O1-main-00/06 and ASU-main-00/06, provide additional implementation checks. Pilot and earlier-study cases are excluded from the performance estimate by design; all 44 evaluation cases are retained.

Let $u_i \sim \mathrm{Unif}[-1, 1]$, $\theta_i = \Delta u_i$ and $a_m(\theta) = \exp(\mathrm{j}\pi m \sin\theta)/\sqrt{M}$. Independent vectors $z_i, z_{is} \sim \mathcal{CN}(0, I_M/M)$ define a user-specific component shared across resources and an independent user/resource component:

$$v_{is} = \sqrt{10/11}\, a(\theta_i) + \sqrt{1/11}\big(\sqrt{0.8}\, z_i + \sqrt{0.2}\, z_{is}\big), \qquad d_{is} = v_{is}/\|v_{is}\|.$$

User offsets are uniform on $[-6, 6]$ dB, and resource offsets on $[-3, 3]$ dB. With their sum δ_{is},

$$g_{is} = 10^{(\mathrm{SNR}_0 + \delta_{is})/10}/0.1, \qquad h_{is} = \sqrt{g_{is}}\, d_{is}.$$

The four regimes share the same latent draws within a seed, and inference resamples whole seeds. The pre-normalization LOS-to-diffuse expected power ratio is 10 because both components have unit expected squared norm. After direction normalization and prescribed gain, the vector is not an unmodified Rician amplitude distribution. These physically realizable controlled arrays are not measured or standard-compliant deployments. Physical noise can be restored with $\sigma^2 = 10^{(-174+7-30)/10}(50\times10^6/512)$ W. No FFT correction is applied to this directly defined resource model.

The independent checker reconstructs exact rational gains and squared correlations from the stored binary64 channel components. Zero or nonfinite generated norms invalidate an input without resampling. The generator records PCG64, package versions, draw order, latent offsets and channel/coefficient hashes. The displayed diffuse normalization was used for all generated inputs.

S8.2 Algorithms and charged work

T uses the numerical power-transfer refinement described in the main article. W retains the matching/WMMSE updates, eight random starts and uniform start, with the common checked anchor added. Its short exchange and outer-iteration limits are replaced by convergence or deadline stopping, subject to a recorded emergency limit. S uses SLSQP with analytic gradients, 10^{-10} function tolerance and a 500-iteration per-call limit; a limited call resumes from its feasible endpoint while time remains. Fu-SCA and Thu-IDE use the power components in Section S7. T, S, Fu-SCA and Thu-IDE share the permitted multistart and four-user exchange search, although their completed searches can differ.

Shared preprocessing has a 4-s allowance. It first checks a uniform fallback and then attempts Grid, with native ILP/LP limits of at most 1.25 s each and time reserved for checking. Every arm receives the best timely checked Grid/uniform anchor and is charged the full 4 s. Each arm then has 16 actual seconds: 11 s for startup, imports, loading and search, followed by 5 s for independent checking and saving results. The total charged allowance is therefore 20 s. Arm order rotates deterministically by input index. Arms run serially with single-threaded numerical libraries; a candidate may be checked while its search process continues within the same elapsed-time allowance.

Only independently checked values saved by the deadline can improve the retained result. At a deadline, the last such value is retained. A missing checked fallback, invalid input, unexpected computation error, changed source or a new lower bound exceeding the verified upper bound for an earlier case stops the study.

Failed and unrun executions remain in the record, and a feasible fallback does not conceal a computation error.

The prespecified allocation totals 6848 s within a 7200-s limit: 260 method executions at 20 s each, 208 s of shared preprocessing, 480 s for correctness checks, 480 s for references, 180 s for generation/analysis and 300 s of contingency. Shared preprocessing is conservatively charged to every arm and also counted once per input in the total. This allocation determines the study size; unused time does not replace an observed case.

S8.3 Endpoints and uncertainty

For each evaluation case, let T denote the transfer arm's checked gross SE and $B = \max\{\mathrm{SE}_{\mathrm{Fu}}, \mathrm{SE}_{\mathrm{Thu}}\}$. This envelope combines two separately budgeted 20-s methods. For each of the eleven seeds, we average $T - B$ and $(T - B)/B$ over the four regimes. The prespecified substantial-effect criterion requires both 95% paired whole-seed bootstrap lower endpoints to exceed 0.10 bit/s/Hz and 5%, respectively. These thresholds define substantial improvement for this experiment. PCG64 seed 2026110399 generates 20,000 resamples of eleven whole groups. All 44 cases must be complete and valid; a zero denominator makes the relative effect undefined.

We also report T–W, T–S and contrasts with the envelope of all four baselines. A substantial gain over that envelope requires the same two thresholds and attainment of the primary criterion. The 0.02-bit/s/Hz all-baseline diagnostic is a smaller secondary endpoint. Individual and regime contrasts, Jain index, minimum-user rate, elapsed time and verified-incumbent histories are descriptive. All eleven group effects, 44 case contrasts and leave-one-group-out summaries are retained. Eleven independent groups limit precision and generalization; cohort size follows the computational budget rather than a prospective power calculation.

S8.4 Independent correctness references

The nonorthogonal line kernel is compared against independently constructed rational derivatives and real-root isolation in twelve fixtures: $r \in \{0, 1/2, 1\}$, gains $(1, 1)$ or $(1, 10)$, common initial power 0.1 W, and users in the same or different scheduled pairs. Zero-gain, equal/active-bound, constant-polynomial, degree-trimming, repeated-root, near-real-root, injected-root-failure and budget-slack cases are separate implementation branch tests, not additional independent rational-reference comparisons. The allowed restricted-objective deficit in the twelve reference fixtures is 10^{-6} nat. Numerical root proposals do not acquire exact completeness from passing these finite fixtures.

Eight separate orthogonal $K = 4, S = 2, M = 4$ cases use $h_{is} = ((16 + n_{is})/16)e_i$, with $n_{is} \in \{0, \ldots, 20\}$. The four development seeds, 2026113601–2026113604, precede the four reference seeds held out during development, 2026113701–2026113704. An independent rational water-filling/dual calculation enumerates all six labelled assignments and must enclose the global optimum to 10^{-6} nat. Each 60-s reference allowance includes at most 20 s for that calculation, 6 s for each of five diagnostic arms and 10 s for checking and saving results. Feasible heuristic values must not exceed the reference upper bound; their objective deficits are reported. These 6-s diagnostics are separate from the 20-s performance measurements, and this orthogonal subclass does not certify general nonorthogonal eight-user optima.

S8.5 Complete outcomes

All five arms completed on all 44 evaluation cases, with independently checked retained values. The four pilot and four earlier ray-traced cases remain separate implementation checks. Eight orthogonal references (four development and four held out during development) each enumerated all six labelled assignments and attained a certified gap below 10^{-6} nat. Their saved records contain final bounds and assignment counts, but not all per-assignment dual witnesses. Saved-data verification therefore checks consistency and retained feasibility/rate without independently reconstructing all six reference upper bounds.

T did not meet the prespecified substantial-effect criterion against the Fu-SCA/Thu-IDE envelope or against the four-baseline envelope. It also missed the secondary 0.02-bit/s/Hz criterion. The positive contrast with Grid is secondary to these outcomes. Intervals near zero do not establish equivalence because no

equivalence margin was tested. The reproduction material contains all group effects, leave-one-group-out means, verification times and rational allocations.

Table S8: Descriptive means over all 44 cases. Time includes the four-second common-preprocessing charge.

Arm	SE (bit/s/Hz)	Charged s	Minimum-user SE	Jain
Transfer T	10.82590489	15.086	0.980269	0.784150
Matching–WMMSE	10.82612692	15.092	0.980534	0.783389
SLSQP	10.82612801	15.153	0.980290	0.783369
Fu-SCA	10.82592527	15.096	0.980327	0.784258
Thu-IDE	10.82568307	15.158	0.980290	0.783689

All arms obey the same box and sum-power constraints, with mean total power approximately 0.8 W. Memory sampling was unavailable; recorded zero RSS values represent missing measurements. Inner search may return slightly after its cooperative stopping time while remaining inside the hard result-saving allowance. Reported times include final checking and process cleanup and do not establish real-time operation.

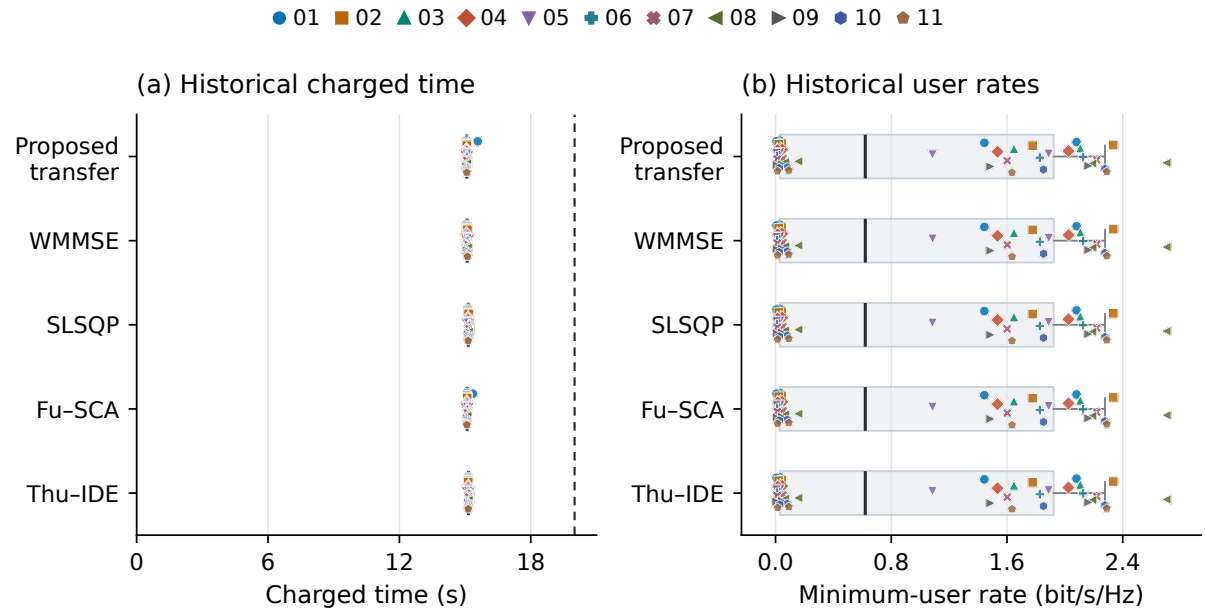


Figure S10: All 220 joint-search charged durations and minimum-user rates, by method and seed group. Boxes show medians, interquartile ranges and 5th–95th percentile whiskers; all case points are included. The dashed line marks the 20-s allowance. Proposed transfer denotes T; Fu–SCA [12] and Thu–IDE [13] are the component adaptations in Section S7.

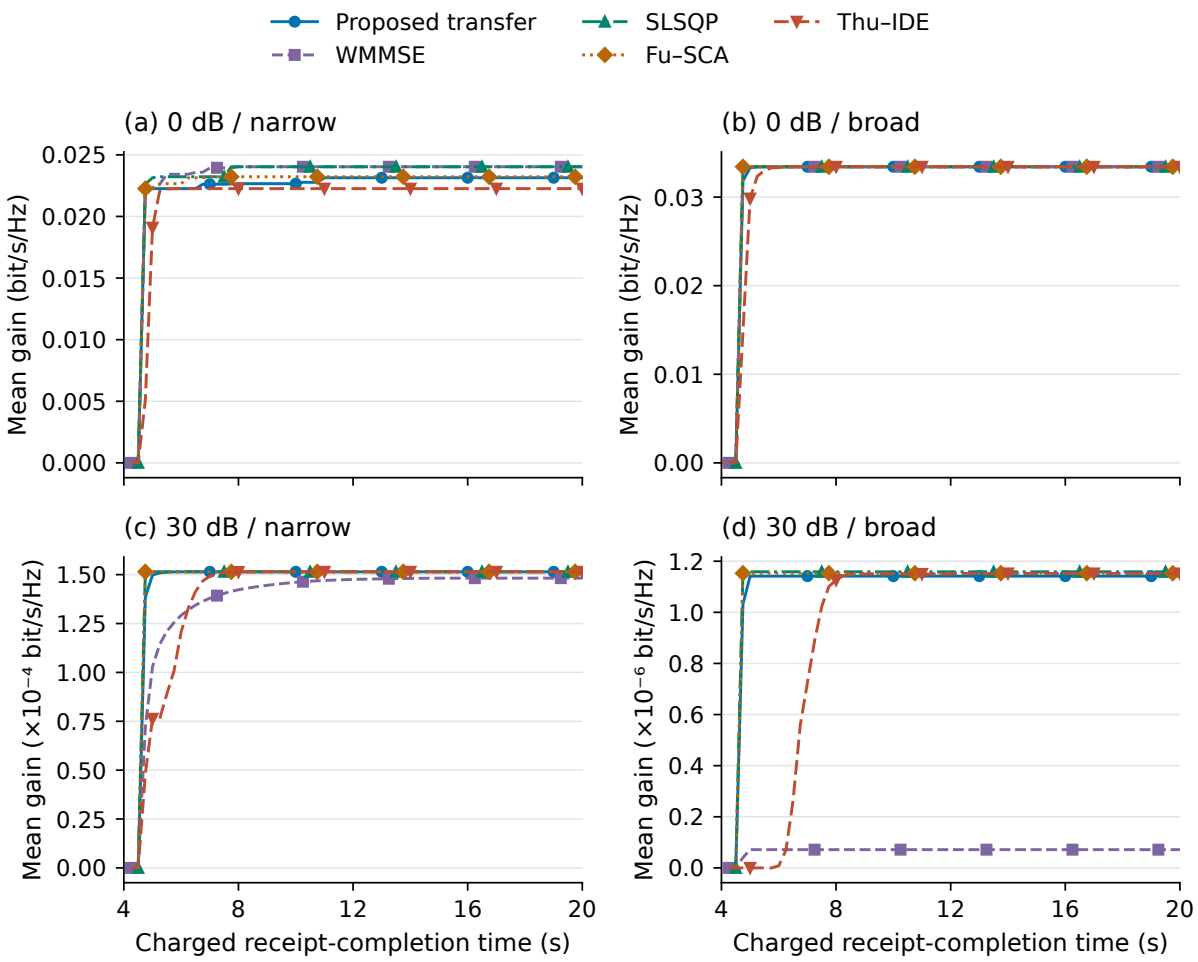


Figure S11: Mean verified-incumbent gain over the common anchor by regime and method. Each trajectory begins at its first completed verification record; at each plotted time, the mean uses the available verified gains. Per-time counts and all 220 case trajectories are supplied with the figure data. Proposed transfer denotes T; Fu–SCA [12] and Thu–IDE [13] use the component adaptations in Section S7.

Table S9: All controlled-array case rates (bit/s/Hz). LN/LB denote 0 dB narrow/broad, and HN/HB denote 30 dB narrow/broad. Seed suffixes 01–11 correspond to 2026113401–2026113411. The final column is T minus the Fu-SCA/Thu-IDE envelope. Rates have six decimal places (half-unit threshold 5×10^{-7}), and contrasts have three significant digits. Statistical calculations use the archived unrounded values.

Case	T	W	S	Fu-SCA	Thu-IDE	T–envelope
01-LN	2.525385	2.525385	2.525385	2.525385	2.525385	1.95×10^{-9}
01-LB	2.849497	2.849497	2.849497	2.849497	2.849497	-3.16×10^{-9}
01-HN	16.587643	16.587638	16.587643	16.587643	16.587643	-1.31×10^{-8}
01-HB	20.386591	20.386589	20.386591	20.386591	20.386591	-1.68×10^{-9}
02-LN	1.713109	1.713109	1.713109	1.713109	1.713109	7.48×10^{-11}
02-LB	2.103551	2.103551	2.103551	2.103551	2.103551	3.57×10^{-10}
02-HN	15.993150	15.993148	15.993150	15.993150	15.993150	-2.84×10^{-8}

Case	T	W	S	Fu-SCA	Thu-IDE	T–envelope
02-HB	19.873274	19.873274	19.873274	19.873274	19.873274	-8.46×10^{-9}
03-LN	2.141727	2.141727	2.141727	2.141727	2.141727	1.40×10^{-9}
03-LB	2.525361	2.525361	2.525361	2.525361	2.525361	-3.21×10^{-9}
03-HN	16.815143	16.815141	16.815143	16.815143	16.815143	-2.08×10^{-8}
03-HB	20.092731	20.092730	20.092731	20.092731	20.092731	9.08×10^{-10}
04-LN	2.489984	2.490880	2.490880	2.490880	2.485933	-8.96×10^{-4}
04-LB	2.832161	2.832161	2.832161	2.832161	2.832161	-1.35×10^{-9}
04-HN	16.603188	16.603182	16.603188	16.603188	16.603188	-1.05×10^{-8}
04-HB	20.333803	20.333801	20.333803	20.333803	20.333803	-4.28×10^{-8}
05-LN	2.722501	2.722501	2.722501	2.722501	2.718558	1.69×10^{-9}
05-LB	3.039559	3.039559	3.039559	3.039559	3.039559	5.21×10^{-10}
05-HN	15.714362	15.714359	15.714362	15.714362	15.714362	-1.52×10^{-8}
05-HB	20.475352	20.475349	20.475352	20.475352	20.475352	-1.72×10^{-8}
06-LN	2.574284	2.583204	2.583204	2.574284	2.574284	6.42×10^{-10}
06-LB	2.988643	2.988643	2.988643	2.988643	2.988643	-2.74×10^{-9}
06-HN	17.489821	17.489817	17.489821	17.489821	17.489821	2.96×10^{-9}
06-HB	20.913153	20.913152	20.913153	20.913153	20.913153	-3.63×10^{-8}
07-LN	2.751530	2.751530	2.751530	2.751530	2.751530	7.90×10^{-10}
07-LB	3.250964	3.250964	3.250964	3.250964	3.250909	1.86×10^{-9}
07-HN	17.394107	17.394104	17.394107	17.394107	17.394107	-3.81×10^{-9}
07-HB	21.254763	21.254762	21.254763	21.254763	21.254763	-1.35×10^{-8}
08-LN	3.292327	3.292327	3.292327	3.292327	3.292327	1.49×10^{-9}
08-LB	4.378011	4.378011	4.378011	4.378011	4.378011	-4.64×10^{-9}
08-HN	20.505359	20.505358	20.505359	20.505359	20.505359	-5.66×10^{-9}
08-HB	23.443155	23.443155	23.443155	23.443155	23.443155	0.00×10^{0}
09-LN	2.398615	2.398615	2.398615	2.398615	2.398149	2.43×10^{-9}
09-LB	2.723511	2.723511	2.723511	2.723511	2.723511	2.80×10^{-10}
09-HN	15.887178	15.887174	15.887177	15.887177	15.887177	3.27×10^{-9}
09-HB	20.243684	20.243682	20.243684	20.243684	20.243684	1.28×10^{-8}
10-LN	2.286069	2.286069	2.286069	2.286069	2.284823	9.09×10^{-10}
10-LB	2.715677	2.715677	2.715677	2.715677	2.715677	1.18×10^{-9}
10-HN	17.649794	17.649790	17.649795	17.649794	17.649794	-3.69×10^{-9}
10-HB	20.652621	20.652620	20.652621	20.652621	20.652621	-1.44×10^{-8}
11-LN	2.401057	2.401057	2.401057	2.401057	2.401057	1.38×10^{-9}
11-LB	3.047917	3.047917	3.047917	3.047917	3.047917	2.83×10^{-10}
11-HN	16.989406	16.989403	16.989406	16.989406	16.989406	-2.45×10^{-8}
11-HB	21.290098	21.290097	21.290098	21.290098	21.290098	-4.06×10^{-8}

Table S10: Prespecified group-bootstrap effects. "Modern" denotes the Fu-SCA/Thu-IDE envelope; relative effects are percentages. Intervals use 20,000 whole-seed resamples with the fixed PCG64 seed.

Contrast	Mean	95% interval
T–modern (bit/s/Hz)	$-2.03777e-05$	$[-6.11226e-05, -3.36369e-09]$
T–modern (%)	−0.000817866	$[-0.00245355, -9.96724e-09]$
T–four-baseline (bit/s/Hz)	−0.000223118	$[-0.000628597, -6.73324e-09]$
T–four-baseline (%)	−0.00866623	$[-0.0243629, -4.31594e-08]$
T–W	−0.000222029	$[-0.000627611, 1.08601e-06]$
T–S	−0.000223118	$[-0.000628596, -6.42967e-09]$
T–Fu-SCA	$-2.03759e-05$	$[-6.11211e-05, -1.82936e-09]$

Contrast	Mean	95% interval
T–Thu-IDE	0.000221825	$[2.49164e-05, 0.000464791]$

S8.6 Completed search work

Table S11 summarizes the saved diagnostics for all 44 cases per arm. A completed start includes its exchange search where applicable. For W, exchanges begin after the initial ten starts (anchor, uniform and eight random) have been visited. The completed-start count therefore differs from the number of initial power problems solved.

A completed exchange trial includes power refinement and candidate checking; it need not improve the incumbent or visit a distinct assignment. Attempted exchanges are counted before refinement. Recorded power calls include start, exchange and matching-start calls; the last type combines matching and power for W. These counts do not measure comparable inner iterations or converged solves. A deadline can prevent the final call record even when that call produced verified candidates. Total initiated calls and unfinished-call generation counts are therefore unavailable.

Table S11: Completed search work, median [minimum, maximum] over 44 cases per arm. "Calls" counts recorded power-bearing calls. All displayed counters and arrays are present in every case.

Arm	Completed starts	Completed exchange trials	Recorded calls	Accepted exchanges
T	1 [0,2]	41.5 [19,86]	43.5 [20,88]	2 [0,6]
W	9.5 [2,23]	86.5 [0,420]	97 [3,444]	0 [0,2]
S	8 [5,17]	824 [436,1436]	835 [443,1454]	59.5 [32,107]
Fu-SCA	3 [1,9]	279.5 [100,784]	284 [103,794]	19.5 [6,59]
Thu-IDE	0 [0,0]	0 [0,0]	0 [0,0]	0 [0,0]

For attempted exchanges, the corresponding medians [ranges] are T: 42.5 [20,87], W: 87 [0,421], S: 825 [437,1437], Fu-SCA: 280.5 [101,785], and Thu-IDE: 0 [0,0]. Thu-IDE starts one initial assignment in every case, completes no outer start and attempts no exchange. Its empty call-record arrays result from the post-call deadline checks; the method still searches and produces verified candidates. W completes no exchange trial in 22/44 cases; T, S and Fu-SCA complete exchanges in every case.

No case is excluded from the rate analysis. Missing diagnostic fields would be marked unavailable rather than zero; all displayed fields are present, although the final unrecorded call cannot be reconstructed. The extraction checks compressed and uncompressed diagnostic hashes and agreement with the result counters. The reproduction material includes the extraction script, 220-row export, summary and input hashes.

Every arm uses the same 20-s ceiling and the prespecified remaining-time policy for power calls. The Fu-SCA/Thu-IDE envelope still combines two separately budgeted procedures. Because completed assignment searches differ, these rate contrasts compare end-to-end adapted methods. The fixed-assignment study in Section S9 isolates their power updates on the same 44 inputs; it does not increase the number of independent cases or seed groups.

S8.7 Influence of individual seed groups

Figure S12 shows all eleven group effects from the original joint-search comparison. Group 2026113404 contributes -2.240983×10^{-4} bit/s/Hz and accounts for 99.975% of the signed overall sum. Omitting it solely as a diagnostic changes the mean to -5.5969×10^{-9} bit/s/Hz; it remains in every prespecified endpoint and bootstrap. The median group effect is -5.4314×10^{-9} bit/s/Hz. These results show the sensitivity of tiny mean differences while leaving the failed substantial-effect endpoint unchanged.

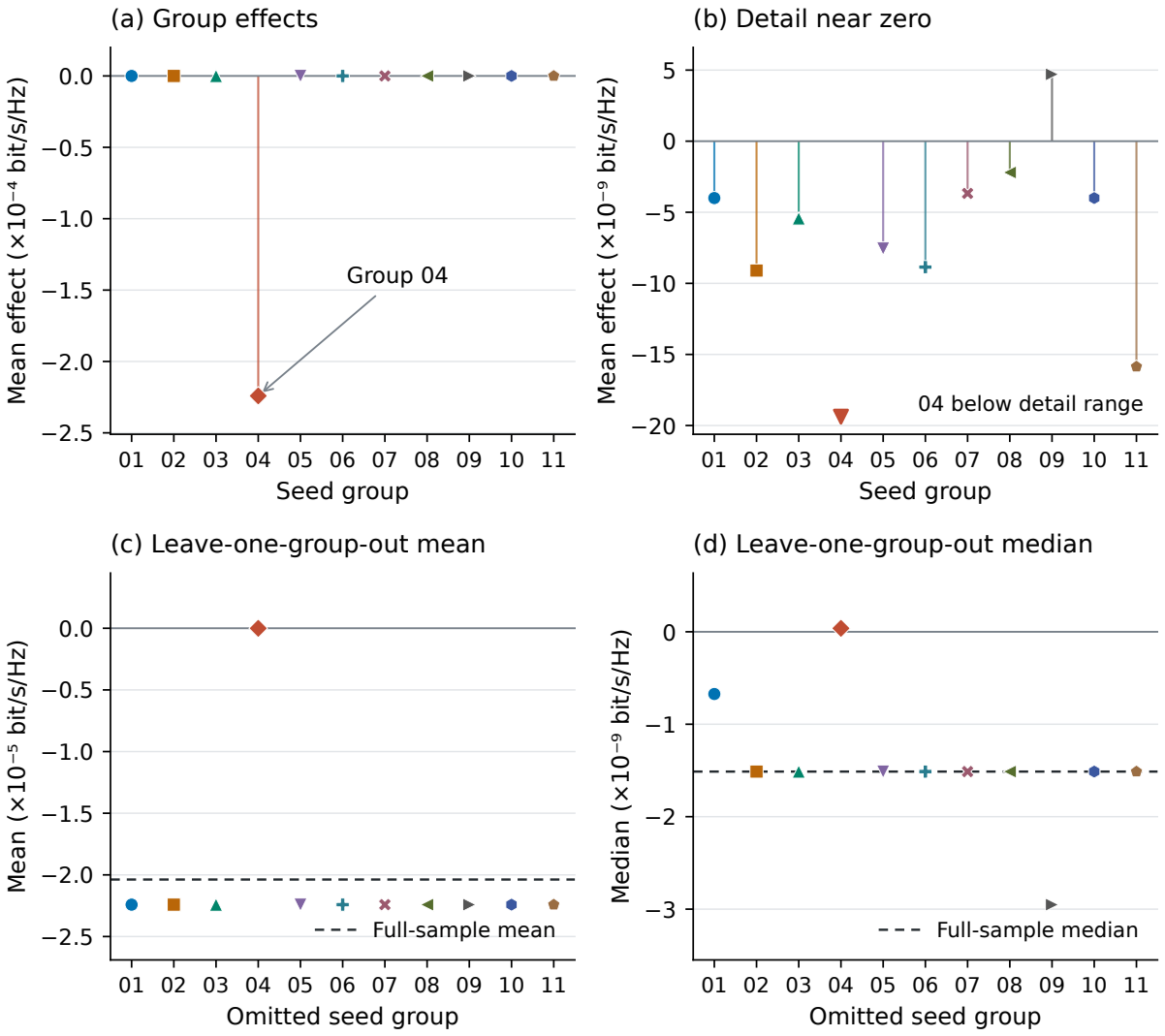


Figure S12: Joint-search Proposed transfer minus the per-case Fu–SCA [12]/Thu–IDE [13] envelope. Panels (a,b) show all eleven group means and the near-zero detail; panels (c,d) show leave-one-group-out means and medians, with dashed full-sample references. Group 04 (seed 2026113404) remains in the primary analysis and is indicated below the detail range.

S9 Fixed-assignment comparison and small-network global references

S9.1 Common assignments and initial powers

The fixed-assignment study reuses the same 44 controlled-array cases and 44 Grid anchors from Section S8. The independent sampling units remain the same eleven seed groups. Within each case, all five arms start from the same assignment and exact rational powers. Only the power-refinement routines are called; assignments remain fixed throughout, with no matching, assignment exchanges or multistart search. Fu-SCA and Thu-IDE use the component adaptations in Section S7.

Each arm receives a five-second cooperative refinement limit. Initial and final independent checks use the integer-log verifier outside this timer. If the final lower enclosure falls below the initial lower enclosure, the common start is retained. This occurred in 18 executions: T/W/S/Fu-SCA/Thu-IDE counts 1/7/1/1/8. Two Thu-IDE calls exceeded the cooperative limit by at most 0.016 s. All 220 executions had valid initial and final schedules, and every assignment remained unchanged. The saved results include proposed and retained powers, verification records, search diagnostics and timing.

An initial development run stopped after a checked decrease, leading to the retention rule above before the complete evaluation. Its partial results are not pooled with the completed study. A later correction to the leave-one-seed-out calculation changed only postprocessing; all 220 completed execution records remained unchanged. Both corrections are documented with the saved evidence.

S9.2 Verified rate gains

Table S12 reports gross occupied-resource SE, $L/(S\ln 2)$ with $S = 4$. The anchor and constraints are identical within each case. Reported refinement times exclude independent verification. We show individual Fu-SCA and Thu-IDE contrasts because their envelope requires both runs and twice the single-arm refinement allowance.

Table S12: Fixed-assignment outcomes over all 44 controlled-array cases.

Arm	Mean gain (bit/s/Hz)	Mean final SE	Mean refinement s
Transfer T	0.01395755	10.82568431	0.9415
WMMSE W	0.01395629	10.82568304	0.8196
SLSQP S	0.01395756	10.82568431	0.0336
Fu-SCA	0.01395756	10.82568431	0.1417
Thu-IDE	0.01394897	10.82567572	5.0007

Table S13 uses 20,000 whole-seed bootstrap resamples with seed 2026110399. The four regimes of each seed remain together. Signed differences are retained at the displayed precision; their small magnitude alone does not establish equivalence.

Table S13: Paired fixed-assignment contrasts in bit/s/Hz.

Contrast	Mean	95% whole-seed interval
T − Fu-SCA	-4.632×10^{-9}	$[-7.504, -1.785] \times 10^{-9}$
T − Thu-IDE	8.582×10^{-6}	$[4.920, 12.996] \times 10^{-6}$
T − SLSQP	-7.749×10^{-9}	$[-10.076, -5.136] \times 10^{-9}$
Fu-SCA − Thu-IDE	8.586×10^{-6}	$[4.925, 13.000] \times 10^{-6}$

T has a higher retained value than Thu-IDE in 43 cases and ties in one, but its mean advantage is far below the preceding study's 0.10-bit/s/Hz substantial-effect threshold. Against Fu-SCA, T has 25 wins, one tie and 18 losses, with a negative mean at numerical-tolerance scale. SLSQP is marginally ahead of both T and Fu-SCA. Fixing the assignment therefore removes the assignment-search imbalance without revealing a practically substantial rate advantage. Stopping reasons are coordinate tolerance for T in 44/44 cases,

solver convergence for S in 44/44, numerical stall for Fu-SCA in 44/44, objective tolerance for W in 42/44 and deadline for Thu-IDE in 44/44.

S9.3 Nonorthogonal four-user global references

Two deterministic physical $K = 4, S = 2, M = 4$ nonorthogonal cases, with seeds 20261106 and 20261107, supplement the orthogonal references. Spatial refinement enumerates all six resource-labelled assignments. Each assignment has a saved root, feasible assignment-local schedule and self-contained upper-bound proof. Independent verification clears checker caches before every witness and reconstructs all six roots, feasibility and bounds without importing the optimization program. Both cases attain the separate 10^{-3}-nat target, with certified global gaps of 3.4864×10^{-10} and 3.1963×10^{-10} nat. The small gaps bound the best feasible incumbent's regret; they do not establish a uniquely optimal discrete assignment. Each bundle records split and time limits and witness hashes.

For a limited heuristic check, every arm starts at uniform feasible power on the assignment of the producer's best feasible incumbent. Table S14 reports the difference between the independently verified global upper bound and each heuristic lower bound. This conservatively bounds the objective deficit rather than measuring exact regret. SLSQP has the smallest value in both fixtures, while Thu-IDE has a larger deficit in case B. These two four-user examples establish neither general eight-user optimality nor a runtime scaling law.

Table S14: Certified global-upper-minus-heuristic-lower deficits (bit/s/Hz).

Arm	Physical case A	Physical case B
T	1.440×10^{-9}	7.912×10^{-10}
W	1.067×10^{-9}	2.054×10^{-9}
S	2.992×10^{-11}	4.456×10^{-11}
Fu-SCA	1.283×10^{-9}	2.349×10^{-9}
Thu-IDE	4.360×10^{-10}	5.311×10^{-5}

The reproduction material includes the generator, spatial optimizer, independent verifier and fixed-assignment evaluator. The nonorthogonal witnesses provide replayable global bounds at $K = 4$; comparison with a strong global solver on larger networks remains unresolved.

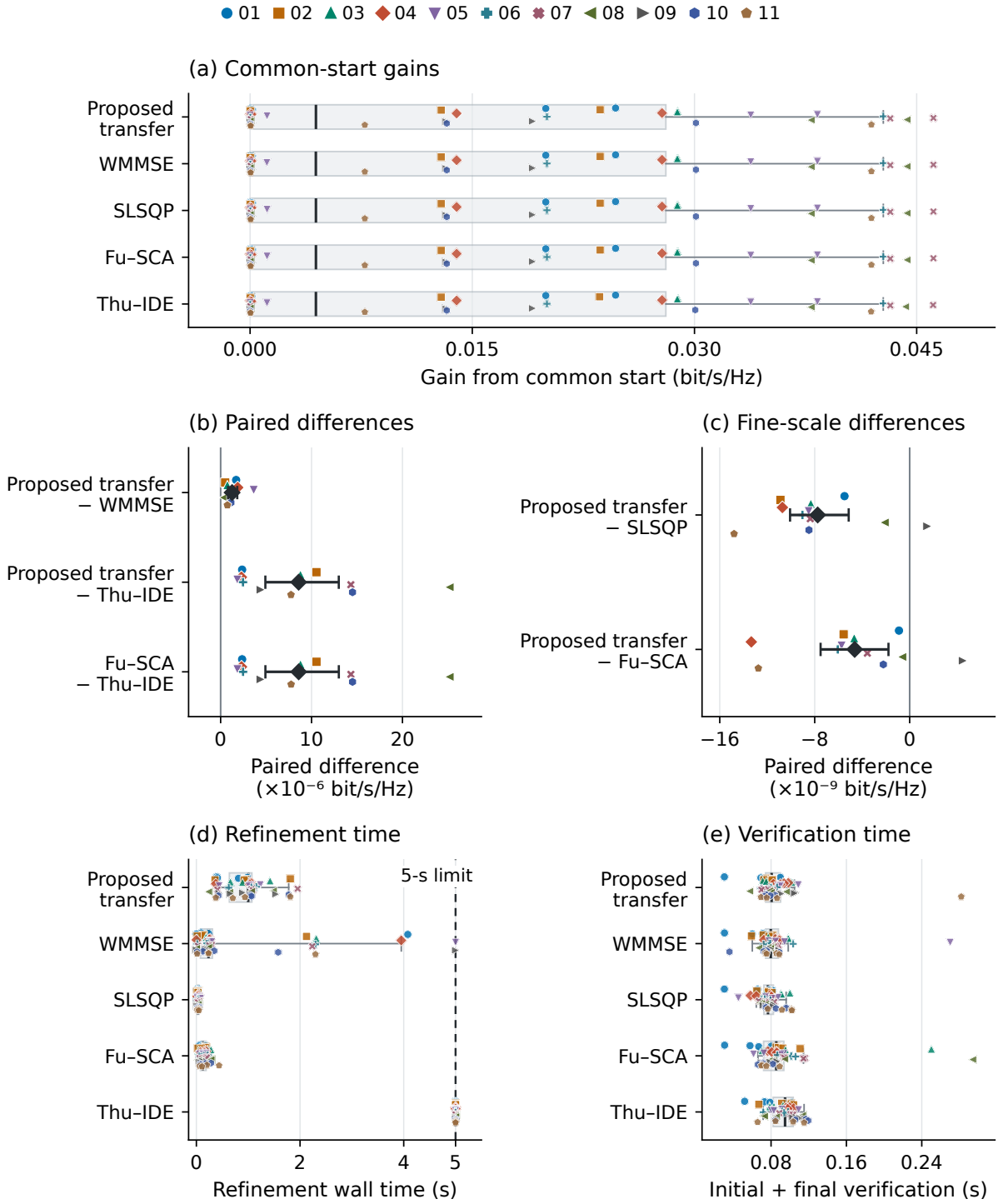


Figure S13: Common-start gains, five paired contrasts, refinement times and separate initial-plus-final verification times for all 44 cases. Proposed transfer denotes T; Fu–SCA [12] and Thu–IDE [13] use the adaptations in Section S7. Diamonds and intervals show paired means and 95% whole-seed bootstrap intervals.

S10 Data, code and reproduction

The accompanying release contains editable paper sources in `source/` and scientific records in `evidence/`. These include vector figures, scalar outcomes, stored channel instances, rational allocations, proof partitions, verification records and analysis programs. The release guide maps each study to its preserved bundle and gives the corresponding working directory and commands. The complete release, including the 3GPP CDL generator and case files, is provided to reviewers with the submission; the public repository will be released upon acceptance.

S10.1 Channel inputs and computational environment

Ray-traced channels use DeepMIMO [7] with propagation inputs generated by Remcom Wireless InSite. Raw scene archives and generated channel pools are obtained separately. The required inputs are the O1_60 scene and the legacy ASU_Campus1 scene, also identified by the DeepMIMO migration guide. Regeneration must use these scenes and the recorded parameter arrays. Stored case coefficients are sufficient for verifying the reported bounds without the raw scenes.

One-AP optimization and checking used Python 3.12.4, NumPy 2.3.4, SciPy 1.16.3 and SymPy 1.14.0. Channel generation used DeepMIMOv3 0.2.9, Python 3.9.25 and NumPy 1.26.4. The recorded host is an Intel Core i7-14700HX Windows system with 20 physical/28 logical cores and approximately 64 GiB RAM. OpenBLAS, OpenMP and MKL thread variables were set to one before numerical imports. The host was not dedicated, and the exact historical BLAS build and separate HiGHS version were not recorded. SciPy supplied `milp` and `linprog(method='highs')`.

Generation metadata include installed-source and input-file hashes, full parameter arrays, selected pool offsets, FFT indices and case/channel hashes. Pool offsets index the generated array rather than physical scenario rows. A generator commit and semantic scene-release tags were not recorded. The original O1 archive digest is also unavailable, although its actual input MAT files were hashed; the ASU archive digest and acquisition URL were retained. These records, together with the stored cases, define the reproducible inputs. The controlled-array and fixed-assignment bundles contain their own generator and dependency records.

S10.2 Verification and analysis from saved results

Saved-data programs regenerate figures, recompute statistical summaries and independently check retained schedules and available global proofs. These operations do not repeat channel generation or optimization, and their elapsed times do not reproduce historical performance measurements on another host. Each output directory or report path must be new.

For the original one-AP study, verification covers 128 executions and 590 records of timely completed checks. The corrected study has 128 executions and 700 such records. Identical case/schedule/proof inputs may reuse a reconstructed mathematical result only after checking each record's integrity and timing; the corrected replay reused 417 repeated inputs. The independent verifier imports neither the optimization program nor a numerical optimizer and uses the arithmetic in Section S5. Its 76.4416-s proof replay was part of the 92.894-s saved-data calculation described in Section S6.

After extracting the preserved one-AP scientific bundle to a separate directory, run the following command from that bundle's root:

```
python reproduction/reproduce.py --study corrected --mode certificates --output replay
```

The release guide gives the nested extraction path and identifies the checker that rejects missing planned executions. Verification is limited to the proof witnesses actually retained in each bundle.

The controlled-array replay checks all planned common and method records, verifies retained feasibility and rate, and recomputes the eleven group effects and 20,000-resample bootstrap summaries. Its default numerical scope covers retained schedules; an optional all-timely scope also checks each distinct timely candidate. The eight orthogonal references retain final bounds and assignment counts but lack complete per-assignment dual witnesses, so replay checks their consistency without re-proving the global upper bounds.

The fixed-assignment replay checks 220 executions and 440 independent initial/final numerical validations, along with file integrity and paired summaries. These records concern the same 44 controlled-array

inputs. Each nonorthogonal four-user bundle supplies all six assignment witnesses, which are reconstructed independently without importing the optimizer. Exact script paths, fresh-report arguments and reproduction allowances are listed in the bundle's guide. The analysis and figure programs read the preserved outcomes; changes to presentation do not create additional observations.

The finite-box/integral study in Section S11 retains all 32 inputs, both methods' witnesses, exact result rows and descriptive analyses. Saved-data replay reconstructs 64 new/generic bounds and checks 84 historical feasible allocations. The release guide supplies explicit archive paths and a fresh output directory for a portable working copy. Replay verifies mathematical endpoints; it does not remeasure the original construction times.

S10.3 Earlier studies retained in the archive

The earlier scientific bundles preserve complete methods, per-case tables, rate normalizations and unsuccessful executions for the studies summarized below. They use different inputs or decision problems and are not pooled with the evaluations in Sections S6–S9.

Imperfect-CSI development. The learned pilot-stopping policy gave no gain over the best fixed length selected during development in any of nine settings. In one 80-path setting, the sole early stop violated the sufficient oracle comparison: the marginal count was 1/80, whereas the count conditional on stopping was 1/1. Marginal split-conformal coverage under exchangeability therefore did not establish conditional safety of the observed stops. In the 32-network comparison with equal acquisition cost, all four contexts' intervals included zero and the mean joint-minus-weakest differences were negative. Certificate-based early termination activated in zero cases. The finite-grid positive-part opportunity was 0.5317%, below the prespecified 1% continuation threshold; in 23/32 cases the best action could not repay the added pilot symbol. The corresponding numerical LP opportunity bound was 0.5918%. These diagnostics concern the tested finite family. They provide neither a population impossibility result nor a guarantee for continuous power under channel-estimation error.

Fixed-estimator CSI follow-up. The separate 120-network study found a CNN-minus-LMMSE mean of -0.023798 bit/s/Hz at four pilots, with 95% Student-t interval $[-0.049732, 0.002137]$ and paired-bootstrap interval $[-0.049089, 0.002429]$. It established neither superiority nor equivalence. Its normalized channel/noise units differ from the physical-watt experiment, and eight pilots consume twice the pilot energy of four. All 1,440 evaluations and complete minimum-rate and estimation-error tables remain in the archive. A nominal-channel optimization bound does not certify the actual rate under channel-estimation error.

CPU/GPU and minimum-power studies. Four initial GPU attempts reached their 40-s limits. After correcting the sparse-matrix representation, GPU coefficient construction was faster only at the largest tested size, while complete GPU LP optimization showed no speedup. The sixteen- and thirty-two-user final iterates were infeasible. Numerical grid-dual bounds in that study have floating-point qualifications and do not replace the continuous certificates. The minimum-power study used 95%-throughput targets, so its reductions do not establish equal-throughput power savings. Separate fixed-schedule uniform scaling already attained its own reduced-rate targets, but differing schedules and targets prevent a matched comparison. Fixed-schedule maximum-rate gains from SLSQP and adapted I-WMMSE were negligible on those inputs.

Four-AP scheduling study. All 176 executions are preserved, including storage errors and incomplete pricing. Full met the original operational target on 5/24 primary cases versus 0/24 for the generic reference, with conservative 95% difference interval $[-0.1070655, 0.4506892]$. Requiring normal process return reduced Full to 3/24 and gave $[-0.1464695, 0.3523712]$. Neither interval establishes superiority. Seven of eight scale pricing runs were incomplete, and no scale case attained the 0.01-nat target. Reported minimum-user rates and Jain indices are descriptive because the design has no QoS constraint. Missing historical proof trees cannot be recovered by the current verifier.

S10.4 Source identification

Metadata supplied with the evidence identify inspected publications, manuscript editions and technical locators. This includes the source-specific comparisons underlying the main related-work discussion and the component adaptations in Section S7. Publication metadata are distinguished from the edition used for equation-level implementation. Bibliographic source texts and raw propagation archives are not redistributed. Historical filenames are preserved in the archive guides so that commands continue to resolve to the original scientific records.

S11 Finite-box curvature and integral scheduling bounds

We prove Propositions 3 and 4. All logarithms are natural. Rectangle endpoints and channel gains are finite, and $r \in [0,1]$ denotes squared normalized channel correlation. The network multiplier $\lambda \geq 0$ prices physical power.

S11.1 Scaled Hessian and finite-box extrema

Let $\mathcal{B} = [\ell_x, u_x] \times [\ell_y, u_y] \subset \mathbb{R}_+^2$, $a = 1 - r$, and $q = 2\ln d - \ln(1+x) - \ln(1+y)$. Define

$$\begin{gathered} d = 1 + x + y + axy, \qquad u = \frac{(1+x)(1+ay)}{d}, \qquad v = \frac{(1+y)(1+ax)}{d}, \\ w = \frac{r(1+x)(1+y)}{d^2}, \qquad J = \operatorname{diag}\left((1+x)^{-1}, (1+y)^{-1}\right). \end{gathered} \tag{B1}$$

Since $d \geq 1$, all quantities are defined on the closed quadrant. The identities $d_x = 1 + ay$, $d_y = 1 + ax$ and $ad - (1+ax)(1+ay) = -r$ give

$$\begin{gathered} q_{xx} = (1+x)^{-2} - 2(1+ay)^2/d^2, \qquad q_{yy} = (1+y)^{-2} - 2(1+ax)^2/d^2, \\ q_{xy} = -2r/d^2, \qquad \nabla^2 q = J \begin{pmatrix} 1-2u^2 & -2w \\ -2w & 1-2v^2 \end{pmatrix} J. \end{gathered} \tag{B2}$$

The matrix J is invertible. Thus $\nabla^2 q \preceq 0$ is equivalent to $2u^2 - 1 \geq 0$, $2v^2 - 1 \geq 0$ and $(2u^2-1)(2v^2-1) \geq 4w^2$.

Differentiating the scaled entries yields

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{ry(1+ay)}{d^2} \geq 0, \quad \frac{\partial u}{\partial y} = -\frac{r(1+x)}{d^2} \leq 0, \\ \frac{\partial v}{\partial x} &= -\frac{r(1+y)}{d^2} \leq 0, \quad \frac{\partial v}{\partial y} = \frac{rx(1+ax)}{d^2} \geq 0. \end{aligned} \tag{B3}$$

Consequently the exact minima on $\mathcal{B}$ are

$$u \geq u_0 = \frac{(1+\ell_x)(1+au_y)}{d(\ell_x, u_y)}, \qquad v \geq v_0 = \frac{(1+\ell_y)(1+au_x)}{d(u_x, \ell_y)}. \tag{B4}$$

These inequalities include zero endpoints and zero-width intervals.

Write $C_\ell = (1+\ell_x)(1+\ell_y)$, $C_u = (1+u_x)(1+u_y)$ and $\alpha(t) = t/(1+t)$. The factorization

$$\delta(x,y) = 1 - r\alpha(x)\alpha(y), \qquad d = (1+x)(1+y)\delta(x,y), \qquad w = \frac{r}{(1+x)(1+y)\delta(x,y)^2} \tag{B5}$$

provides a second bound in addition to monotonicity of d. Indeed, α is increasing and $0 \leq \alpha(t) < 1$ for finite $t \geq 0$. Hence

$$\begin{gathered} \delta(x,y) \geq \delta_U := 1 - r\frac{u_x}{1+u_x}\frac{u_y}{1+u_y} = \frac{d(u_x,u_y)}{C_u}, \\ \delta_U \geq \frac{1+u_x+u_y}{C_u} > 0. \end{gathered} \tag{B6}$$

Both d_x and d_y are strictly positive. Taking an upper bound on the numerator and a lower bound on the denominator of w gives W_1 below. Taking lower bounds on the two positive denominator factors in (B5) gives W_2:

$$W_1 = \frac{rC_u}{d(\ell_x, \ell_y)^2}, \qquad W_2 = \frac{r}{C_\ell \delta_U^2} = \frac{rC_u^2}{C_\ell d(u_x, u_y)^2}, \tag{B7}$$
$$0 \le w \le w_0 := \min\{W_1, W_2\}.$$

The last expression for W_2 avoids subtracting nearly equal quantities. Every exact denominator is positive, including when $r = 1$. No coordinatewise monotonicity of w is assumed; at $r = 1$, for example, $\partial w/\partial x = (1 + y)(y - 1 - x)/(1 + x + y)^3$ changes sign.

Suppose $A_0 = 2u_0^2 - 1 \ge 0$, $B_0 = 2v_0^2 - 1 \ge 0$, and $A_0 B_0 \ge 4w_0^2$. Positivity of u, v, u_0, v_0 implies

$$\begin{aligned} &2u^2 - 1 \ge A_0 \ge 0, \qquad 2v^2 - 1 \ge B_0 \ge 0, \\ &(2u^2 - 1)(2v^2 - 1) \ge A_0 B_0 \ge 4w_0^2 \ge 4w^2. \end{aligned} \tag{B8}$$

Thus the Hessian is negative semidefinite throughout the box. Restriction to each line segment in this convex set proves concavity, including for collapsed boxes. This proves Proposition 3. The inequalities need not be strict.

S11.2 Whole-quadrant alternative and boundary cases

Theorem 2 supplies the independent whole-quadrant alternative $a^2 \ge 1/2$. A direct verification uses the scaled Hessian above. Exact expansion gives

$$\begin{aligned} E &:= d^2(2u^2 - 1 - 2w) \\ &= (1 + ay)^2 x^2 + 2\{a + 2(2a - 1)y + a(2a - 1)y^2\}x \\ &\quad + (2a - 1) + 2(3a - 2)y + (2a^2 - 1)y^2. \end{aligned} \tag{B9}$$

For $a \ge 1/\sqrt{2}$, every coefficient is nonnegative and the constant $2a - 1$ is positive. Therefore $2u^2 - 1 > 2w$; exchanging x, y gives $2v^2 - 1 > 2w$. The negative scaled Hessian is strictly diagonally dominant with positive diagonal and hence positive definite at every finite point. Necessity follows from

$$q_{xx}(0, y) = 1 - 2\left(\frac{1 + ay}{1 + y}\right)^2 \longrightarrow 1 - 2a^2. \tag{B10}$$

If $a < 1/\sqrt{2}$, a positive second derivative occurs at finite y and, by continuity, at a nearby interior point.

The finite-box test is sufficient rather than necessary and does not subsume this global alternative. For example, $r = 1/4$ on $[0, 1]^2$ gives $A_0 = B_0 = 17/32$ and $w_0 = 64/225$, so $A_0 B_0 < 4w_0^2$, although Theorem 2 guarantees concavity.

At $r = 0$, $d = (1 + x)(1 + y)$, $q = \ln(1 + x) + \ln(1 + y)$, $u = v = 1$ and $w = W_1 = W_2 = 0$. Every finite box passes. At $r = 1$, $u = (1 + x)/d$, $v = (1 + y)/d$, $w = uv$, and

$$(2u^2 - 1)(2v^2 - 1) - 4w^2 = 1 - 2(u^2 + v^2) < 0, \qquad u + v = 1 + \frac{1}{d} > 1. \tag{B11}$$

The strict inequality follows from $2(u^2 + v^2) \ge (u + v)^2$. Thus the Hessian is indefinite everywhere at finite powers, and no box with positive area is concave at $r = 1$.

For a zero-width interval, only curvature along the remaining direction is relevant. The full-Hessian condition remains sufficient but can reject a concave restriction: $q(0, y) = \ln(1 + y)$ is concave even at $r = 1$. A singleton is trivially concave. These cases may use a one-dimensional bound or the generic construction below. All formulas defining the finite-box test remain well defined.

S11.3 Supporting tangents and complete rectangular pricing

For a physical subbox $\mathcal{B}^p = [\ell_{p_u}, u_{p_u}] \times [\ell_{p_v}, u_{p_v}]$, let $z = (g_u p_u, g_v p_v)$ and $\phi(p) = q(z) - \lambda(p_u + p_v)$. Suppose its normalized image is certified concave. For any $p^0 \in \mathcal{B}^p$, with $z_0 = (g_u p_u^0, g_v p_v^0)$,

$$\begin{aligned} \phi(p) \le T(p) := \phi(p^0) &+ (g_u q_x(z_0) - \lambda)(p_u - p_u^0) \\ &+ (g_v q_y(z_0) - \lambda)(p_v - p_v^0), \\ q_x(z) = \frac{2(1+ay)}{d} - \frac{1}{1+x}, &\qquad q_y(z) = \frac{2(1+ax)}{d} - \frac{1}{1+y}. \end{aligned} \tag{B12}$$

To prove the inequality, restrict q to the segment from z_0 to z. Concavity makes its derivative nonincreasing, so its increment is at most its initial derivative. This argument applies to boundary anchors and collapsed boxes. Subtracting an affine cost preserves the inequality. For an affine upper $T(p) = c + b_u p_u + b_v p_v$, the exact box maximum is

$$\max_{p \in \mathcal{B}^p} T(p) = c + \max\{b_u \ell_{p_u}, b_u u_{p_u}\} + \max\{b_v \ell_{p_v}, b_v u_{p_v}\}. \tag{B13}$$

Thus a positive slope selects the upper endpoint, a negative slope the lower endpoint, and a zero slope either endpoint. The tangent argument permits any signed price; the network budget inequality requires $\lambda \ge 0$.

For positive gains the normalized prices are λ/g_u and λ/g_v. If $g_u = 0$, the rate is $\ln(1 + g_v p_v)$ and the remaining p_u contribution is $-\lambda p_u$. Its maximum occurs at the physical lower endpoint for $\lambda > 0$; any endpoint is optimal for $\lambda = 0$. Setting the undefined normalized channel correlation to zero is consistent with this rate. If both gains vanish, the pair upper can be evaluated exactly as $-\lambda(\ell_{p_u} + \ell_{p_v})$. The minimum power required for service is still counted, and no division by a zero gain is needed.

For boxes without a concavity certificate, put $f = 2\ln d$ and $g(t) = -\ln(1+t)$. The generic decomposition $q = f + g(x) + g(y)$ is valid for every $r \in [0, 1]$. Its first term satisfies

$$\nabla^2 f = -\frac{2}{d^2} \begin{pmatrix} (1+ay)^2 & r \\ r & (1+ax)^2 \end{pmatrix}, \qquad \big((1+ax)(1+ay)\big)^2 - r^2 \ge 1 - r^2 \ge 0. \tag{B14}$$

The matrix has positive diagonal, so f is concave and its tangent is an upper. The convex function g lies below its chord

$$\operatorname{ch}_{[\ell,u]} g(t) = \begin{cases} \dfrac{(u-t)g(\ell) + (t-\ell)g(u)}{u - \ell}, & \ell < u, \\ g(\ell), & \ell = u, \end{cases} \qquad t \in [\ell, u]. \tag{B15}$$

The tangent to f plus these two chords, followed by subtraction of the physical power cost, is an affine upper for ϕ. The zero-width case uses the endpoint value directly.

Alternatively, let $c(t) = (1 + at)/(1 + t) = a + r/(1 + t)$. Its derivative is $-r/(1 + t)^2 \le 0$, and $q = \ln(1 + xc(y)) + \ln(1 + yc(x))$. Each term is nondecreasing in its own power and nonincreasing in the interfering power. Hence, for $\lambda \ge 0$,

$$\begin{aligned} Q_{\mathrm{MM}}(\mathcal{B}) &= \ln(1 + u_x c(\ell_y)) + \ln(1 + u_y c(\ell_x)), \\ \phi(p) &\le Q_{\mathrm{MM}}(\mathcal{B}) - \lambda(\ell_{p_u} + \ell_{p_v}). \end{aligned} \tag{B16}$$

Taking the minimum of independently valid upper bounds on a given box preserves validity. An entire-rate tangent is included only when concavity has been established by the finite-box or whole-quadrant condition.

For each pair/resource combination, let the finite collection $\{\mathcal{B}_j^p\}_j$ cover its entire original domain $\mathcal{R} = [p_{\min}, p_{\max}]^2$. If $B_j \ge \sup_{p \in \mathcal{B}_j^p} \phi(p)$, then

$$H_{uvs} := \max_j B_j \ge \max_{p \in \mathcal{R}} \{q_{uvs}(p) - \lambda(p_u + p_v)\}. \tag{B17}$$

Shared faces and overlapping boxes do not affect this argument. Every unfinished region retains a valid upper; a discarded region must be covered by such a bound or by a certified dominance argument. A maximum combines covering boxes, whereas a minimum combines valid upper bounds on the same domain. Coverage includes every pair/resource combination, whether or not it appears in a numerical master. Thus valid rectangular upper bounds establish pricing completeness without requiring stationary-candidate enumeration.

S11.4 Integral scheduling and the subset recursion

Let $\mathcal{Z}_I$ be the set of schedules assigning one unordered pair to each labelled resource and serving every user exactly once, with $K = 2S$. For a common $\lambda \geq 0$, define the exact pair prices and the integral bound by

$$\begin{aligned} h_{uvs}(\lambda) &= \max_{p \in \mathcal{R}} \{q_{uvs}(p) - \lambda(p_u + p_v)\}, \qquad H_{uvs} \geq h_{uvs}(\lambda), \\ U_I(\lambda, H) &= \lambda P + \max_{z \in \mathcal{Z}_I} \sum_{s,u<v} H_{uvs} z_{uvs}. \end{aligned} \tag{B18}$$

The rectangles in this definition are the full original individual-power domains. For any feasible continuous allocation with pairs $e_s = \{u_s, v_s\}$,

$$\begin{aligned} \sum_s q_{e_s,s} &= \lambda \sum_i p_i + \sum_s \{q_{e_s,s} - \lambda(p_{u_s} + p_{v_s})\} \\ &\leq \lambda P + \sum_s H_{e_s,s} \leq U_I(\lambda, H). \end{aligned} \tag{B19}$$

Exactly-once service accounts for every power once. Nonnegativity of λ justifies the budget step. Maximizing over feasible continuous allocations proves Proposition 4. The entries of H can have either sign.

For resources ordered as $1, \ldots, S$, let $D(s, A)$ denote the maximum remaining weight using users A on resources $s, \ldots, S$. Valid states satisfy $|A| = 2(S - s + 1)$, so $s = 1 + (K - |A|)/2$. Their recursion is

$$\begin{aligned} D(s, A) &= \max_{\substack{u<v \\ u,v \in A}} \{H_{uvs} + D(s+1, A \setminus \{u, v\})\}, \\ D(S+1, \varnothing) &= 0, \qquad U_I = \lambda P + D(1, \mathcal{K}). \end{aligned} \tag{B20}$$

Infeasible states have value $-\infty$. Each schedule selects one pair for resource s and a schedule for the resulting child state. Conversely, each transition and valid child schedule construct a valid parent schedule. Induction from the empty state proves exactness. Memoization by A alone is valid because its cardinality determines s.

Every pair in A is considered at the current labelled resource. Restricting that pair to contain the smallest remaining user would omit valid assignments. There are $O(2^K)$ subsets and at most $\binom{K}{2}$ transitions per state, giving $O(K^2 2^K)$ arithmetic operations and $O(2^K)$ memory. These bounds count arithmetic operations rather than bit complexity.

S11.5 Conservative arithmetic and upward rounding

For rational data, the curvature expressions and tangent slopes admit exact rational evaluation. Otherwise, outward enclosures can certify the required signs. In the finite-box test, use nonnegative lower bounds on u_0, v_0 and an upper bound on w_0. An interval lower endpoint below zero may first be replaced by zero, since the true quantity is positive. All three test inequalities must then be verified in the conservative direction. An unresolved inequality selects a generic valid bound. A denominator interval containing zero is resolved before division, using the positive expression for W_2 or another valid upper.

Logarithms require outward enclosures. Affine maximization uses exact rational slopes or outward interval evaluation that accounts for signed displacements from the anchor. An upward-rounded derivative alone is insufficient when its displacement is negative. These steps establish a rational pair upper H_{uvs} before fixed-point rounding; rounding an unverified approximation does not establish pricing coverage.

Fix an integer scale $\Gamma > 0$ and use the same represented price λ throughout. For validated rational H and an exactly evaluated rational product λP, set

$$\widehat{H}_{uvs} = \lceil \Gamma H_{uvs} \rceil, \qquad U_{\mathrm{cert}} = \lambda P + \frac{\widehat{D}}{\Gamma}, \tag{B21}$$

where $\widehat{D}$ is the root value of the complete recurrence with integer weights. The implemented scale is $\Gamma = 2^{40}$. The budget product remains exact; only the pair weights are rounded. Exact integer additions and comparisons can be obtained with arbitrary-precision integers or a proved range bound.

For each schedule, every rounded entry is at least its original value, including for negative entries. Maximization preserves this inequality. Each schedule has exactly S entries, and each ceiling adds less than $1/\Gamma$. Evaluating the rounding error at an integer-weight maximizing schedule therefore gives

$$V_{\mathrm{C}} \le U_I \le U_{\mathrm{cert}}, \qquad 0 \le U_{\mathrm{cert}} - U_I < \frac{S}{\Gamma}. \tag{B22}$$

Final conversion of U_{cert} to a reported decimal or floating-point upper bound is upward. The minimum over complete certified evaluations at different prices remains an upper bound; pair tables at different prices cannot be combined entrywise under one budget-price term without further justification.

S11.6 Configuration-dual comparison and the remaining budget gap

Let $\mathcal{Z}_F$ contain nonnegative pair/resource weights with unit user-incidence sums and unit resource sums. Since $\mathcal{Z}_I \subseteq \mathcal{Z}_F$, the finite incidence LP upper-bounds the integral scheduling maximum. For arbitrary real user prices α_i, set $\beta_s = \max_{u<v}(H_{uvs} - \alpha_u - \alpha_v)$. The inequalities $\alpha_u + \alpha_v + \beta_s \ge H_{uvs}$ give

$$\begin{aligned} U_I(\lambda, H) &\le \lambda P + \max_{z \in \mathcal{Z}_F} \sum_{s,u<v} H_{uvs} z_{uvs} \\ &\le \lambda P + \sum_i \alpha_i + \sum_s \beta_s = U_{\mathrm{dual}}. \end{aligned} \tag{B23}$$

The final quantity is Proposition 1's continuous configuration-dual repair for the same λ, H. With exact pair prices, the pointwise configuration-dual inequalities are equivalent to $\alpha_u + \alpha_v + \beta_s \ge h_{uvs}(\lambda)$. This comparison uses weak duality and does not require strong duality for the continuous configuration LP.

The ordering in (B23) concerns common prices and a common pair-bound table. Different prices, upper-bound accuracies or added rounding slack need not preserve it, and no ordering against a budget-aware spatial bound follows. Moreover, U_I bounds the continuous integral optimum and need not bound the fractional configuration-LP optimum.

Exact pair pricing and integral scheduling still relax the shared budget. To see the remaining obstruction, consider complete schedules with box-feasible power allocations indexed by j, total powers c_j, rates R_j, and weights $\theta_j \ge 0$ satisfying $\sum_j \theta_j = 1$ and $\sum_j \theta_j c_j \le P$. Since each such allocation is included in the inner maximization,

$$\begin{aligned} U_I(\lambda, h(\lambda)) &\ge \lambda P + \sum_j \theta_j (R_j - \lambda c_j) \\ &= \sum_j \theta_j R_j + \lambda \left(P - \sum_j \theta_j c_j \right) \ge \sum_j \theta_j R_j. \end{aligned} \tag{B24}$$

The fixed-pair power-mixture example in Section S3 has feasible average power and average rate strictly above the deterministic optimum. Equation (B24) therefore proves a positive gap even after taking the infimum over $\lambda \ge 0$, with only one possible schedule. Integral scheduling removes the fractional incidence relaxation from each price evaluation; it does not guarantee exact recovery under the shared power budget.

S11.7 Evaluation protocol and numerical parameters

We evaluate the complete rectangular prices in (B17) and the integral recurrence in (B20) on all 32 ray-traced networks from Section S6. Each scene contributes twelve cases with $(K, S) = (8, 4)$ and four with $(K, S) = (16, 8)$. The eight-antenna channels, objective, powers $p_i \in [0.05, 0.2]$ W and budget $P = K/10$ W are unchanged. We retain every weak or zero channel. The benchmark scenes and their channel/curvature diagnostics had been inspected before implementation; the numerical settings were fixed before the reported runs. Algorithm development used a separate earlier O1 input, a reliability fixture and nonorthogonal synthetic fixtures. All 32 benchmark networks remain in this shared-data comparison.

Both methods receive the same saved grid assignment and powers from the levels $\{0.05, 0.1, 0.2\}$ W. The comparator is the unmodified archived configuration-dual repair, supplied the archived user and power prices. It covers every pair/resource box using midpoint tangent–secant bounds, eligible whole-quadrant

tangents and mixed-monotone bounds, then repairs the resource prices. It permits one table with at most 1200 bisections at one common power price. The earlier Full method additionally uses spatial search and has a different timing scope. The comparison reuses the grid solutions without rerunning grid optimization.

We allow 10 s for each producer and a separate 15 s for its checker. Measured producer time includes model construction, price proposals, subdivision, scheduling and feasible-power refinement inside the callable; measured checking time includes reconstruction of the coefficients and bounds. Imports, archive loading, separately verified fallback setup and output I/O are excluded. Each case/method is run once, sequentially, with alternating method order and cleared arithmetic caches. Numerical linear algebra uses one thread. The implementation uses Python 3.12.4, NumPy 2.3.4 and SciPy 1.16.3 on Windows 11.

For the integral method, SLSQP first proposes powers on the grid assignment under a one-second internal limit. We clip and repair proposed powers by exact rational arithmetic and retain the grid lower bound whenever it is larger. A nonnegative physical-power price is proposed from the median interior marginal rate, using K/P if there are no numerical interior coordinates and zero if the proposed allocation has visible budget slack. Interior and slack decisions use 10^{-7} W only for proposal generation; the price is represented on the 2^{-30} grid. L-BFGS-B proposes each box's tangent point. We bisect the widest physical-power interval of the leaf with the largest upper bound in the pair having the largest estimated pricing gap, committing both children together. Numerical stopping rules do not establish a bound.

Table S15: Numerical proposal and work limits for the integral method. All certification decisions use reconstructed rational bounds.

Component	Setting
SLSQP power proposals	At most 120 iterations; objective tolerance 10^{-12}
L-BFGS-B tangent proposals	At most 60 iterations and 30 line-search steps; objective/gradient tolerances $10^{-14}/10^{-10}$
Complete price evaluations	At most three; at most 1200 bisections per table
Subdivision stopping proposal	Largest estimated pair gap at most 10^{-7} nat
Price-iteration stopping	Constructed upper minus retained lower at most 10^{-3} nat, next price change at most 10^{-6} nat/W, or remaining time exhausted
Scheduling time reserve	0.15 s for $K = 8$; 1.6 s for $K = 16$
Integer scheduling weights	$\Gamma = 2^{40}$, with $\widehat{H} = \lceil \Gamma H \rceil$ for signed H
Logarithm enclosure	40 positive atanh-series terms, outward integer rounding at scale 2^{100} and an explicit tail bound
Final accuracy criterion	Exact verified $U - L \leq 0.01$ nat

After each complete price evaluation, we refine powers on its maximizing integral schedule and may update the price. We retain the smallest complete upper and the best feasible lower. The selected price table covers all $S\binom{K}{2}$ original boxes: 112 at $K = 8$ and 960 at $K = 16$. A separately checked interference-free ceiling and grid allocation remain available if construction or checking is incomplete. As in (B21), λP remains exact. Upward rounding of the pair weights adds less than S/Γ relative to the unrounded integral bound for the same table and price.

A separate checker reconstructs gains and correlations from the exact binary64 channel entries, checks the full physical power domains, and recomputes every tangent, secant and mixed bound. The finite-box decisions and tangent derivatives are rational, with the positive, cancellation-safe expression for W_2 in (B7). The checker imports no optimizer or producer; only the integer logarithm-enclosure primitive is shared. Its top-down subset recurrence is separate from the producer's bottom-up implementation. Both consider all pairs for every labelled resource. Input hashes, the represented price, every split and tangent point, integer weights, the scheduling sum and feasible powers are saved. Reconstructing these witnesses rejects omitted or duplicated rows, invalid partitions and altered values. Fifteen mathematical and implementation tests cover the Hessian identities, curvature above the whole-quadrant threshold, zero gains, $r = 1$, signed ceilings, small-network enumeration, labelled resources and the remaining budget duality gap.

S11.8 All-case certificates and computational cost

Let U_G, L_G denote the generic-repair endpoints and U_N, L_N the endpoints delivered by the integral method, including integer rounding and intersection with the interference-free ceiling. In these runs $L_G = L_{\text{grid}}$. We report final gaps $G_G = U_G - L_G$ and $G_N = U_N - L_N$, and separately compute the paired upper-bound reduction

$$\rho = 1 - \frac{U_N - L_{\text{grid}}}{U_G - L_{\text{grid}}}.$$

The common lower in ρ prevents feasible-rate refinement from being counted as upper-bound improvement. Certificate decisions use the exact final gap and the threshold 0.01 nat, not the rounded table entries.

All 32 new price families and integral scheduling calculations completed, and separate reconstruction reproduced all 64 new/generic bound records. There were no producer or checking errors, timeouts, missing cases or exceeded time allowances. The integral method attained the final-gap target on 27/32 cases: 16/16 in O1 and 11/16 in ASU. Generic repair attained it on 0/32. Every integral-method upper was smaller than its generic counterpart. The median paired ρ was 97.82% overall and 66.37% in ASU; the O1 median was 99.999996%. These are descriptive paired results on the fixed two-scene population.

Table S16: All 32 outcomes, grouped by scene and user count. The index j follows the saved input order within each group. Rates and gaps are in nat; t_p, t_v are the integral method's producer and verification times in seconds. The final column records $G_N \leq 0.01$ nat; every generic gap exceeds the target. Values are rounded for display, and gaps were computed before rounding. Exact endpoints and witnesses remain in the accompanying artifact.

j	L_{grid}	L_N	G_G	G_N	t_p	t_v	Target
ASU, $K = 8$							
0	32.208132	32.415753	0.626587	0.001062	1.576	0.594	Yes
1	25.040614	25.433566	0.709034	0.010182	1.584	0.630	No
2	33.415336	33.451649	0.831415	2.91×10^{-9}	1.674	0.625	Yes
3	21.062004	21.235710	0.492455	0.005956	1.617	0.603	Yes
4	44.934721	44.970982	0.375315	1.44×10^{-9}	1.660	0.652	Yes
5	34.580210	34.785698	0.717298	0.001955	1.606	0.596	Yes
6	25.822444	26.044713	0.453630	0.005813	2.942	0.598	Yes
7	29.243989	29.420222	0.821359	0.000268	1.588	0.610	Yes
8	20.467787	20.641252	0.493878	0.004186	1.579	0.590	Yes
9	45.079992	45.288910	0.676289	0.000286	1.620	0.604	Yes
10	27.334578	27.504046	0.698955	0.002265	1.573	0.627	Yes
11	41.048373	41.253550	0.578353	0.000778	1.575	0.588	Yes
ASU, $K = 16$							
0	46.257977	46.742402	3.373034	1.431925	2.582	1.445	No
1	72.381528	72.611743	4.914289	0.543941	7.388	1.588	No
2	45.699366	46.238781	3.677917	1.045500	7.174	1.476	No
3	48.554231	48.947516	3.767553	0.885982	5.190	1.461	No
O1, $K = 8$							
0	77.606612	77.606612	0.736759	2.22×10^{-10}	0.113	0.091	Yes
1	75.003805	75.003805	0.744861	5.45×10^{-11}	0.121	0.089	Yes
2	76.204076	76.204076	0.709135	7.91×10^{-11}	0.121	0.090	Yes
3	74.778471	74.778471	0.708594	1.45×10^{-10}	0.125	0.090	Yes
4	76.718514	76.718514	0.707630	5.36×10^{-11}	0.118	0.089	Yes
5	74.040659	74.040659	0.707631	5.32×10^{-11}	0.115	0.091	Yes
6	76.761765	76.761765	0.708315	2.14×10^{-12}	0.116	0.090	Yes
7	73.306619	73.306619	0.706892	5.43×10^{-11}	0.119	0.091	Yes
8	77.646626	77.646626	0.707592	1.07×10^{-10}	0.120	0.087	Yes
9	77.386773	77.386773	0.708650	5.37×10^{-11}	0.122	0.089	Yes
10	75.594369	75.594370	0.706831	3.00×10^{-10}	0.117	0.088	Yes

j	L_{grid}	L_N	G_G	G_N	t_p	t_v	Target
11	74.425319	74.425319	0.708131	1.58×10^{-10}	0.117	0.089	Yes
O1, $K = 16$							
0	155.844316	155.844316	7.152706	1.86×10^{-10}	1.280	1.046	Yes
1	152.390440	152.390440	7.444041	3.43×10^{-10}	1.238	1.108	Yes
2	155.849359	155.849359	7.150558	8.18×10^{-11}	1.278	1.042	Yes
3	153.135755	153.135756	7.152317	4.27×10^{-10}	1.285	1.048	Yes

The five misses are retained in Table S16: ASU with $K = 8, j = 1$ has gap 0.010182356 nat, and its four $K = 16$ cases have gaps 1.431925, 0.543941, 1.045500 and 0.885982 nat. These residual gaps do not indicate incomplete pair coverage or incomplete scheduling calculations. The overall median final gap decreases from 0.708622 nat for generic repair to 9.35×10^{-10} nat. The ASU medians are 0.703995 and 0.003225 nat, respectively, making the remaining scene dependence explicit. Fig. S14 separates bound tightening, achieved rates and computational cost.

Median producer/checker times are 1.429/0.597 s for the integral method and 1.237/0.463 s for generic repair. The corresponding maxima are 7.388/1.588 s and 1.784/0.871 s. The integral method's median producer-plus-checker times are 0.210 s in O1 and 2.222 s in ASU. We measure time to the final checked result, rather than the earliest possible certificate. Tighter bounds therefore do not establish a general speedup, and the single-run timings do not characterize runtime variability.

S11.9 Received SNR, curvature coverage and achieved rates

We assess the unchanged channel conditions using nominal received SNR $0.1g_{is}$, with the noise normalization already included in g_{is}. This is received SNR before pairing interference, not SINR. Each scene contributes $12(8 \cdot 4) + 4(16 \cdot 8) = 896$ user/resource links. Zero-gain links remain in the distribution at $-\infty$ dB. Table S17 reports pooled link quantiles and exact curvature counts over all candidate pair/resource boxes, not only selected pairs. The box denominator is $12 \cdot 112 + 4 \cdot 960 = 5184$ per scene.

Table S17: All-link received SNR at 0.1 W and full-domain curvature coverage. "Combined" accepts either the whole-quadrant condition or the sufficient finite-box condition.

Quantity	O1	ASU
User/resource links	896	896
Zero-gain links	0	24
SNR 10th percentile (dB)	42.30	7.12
SNR median (dB)	47.24	16.42
SNR 90th percentile (dB)	50.42	32.20
Median squared correlation r	0.8884	0.9481
Combined qualifying boxes	5184/5184	1663/5184
Whole-quadrant qualifying boxes	51/5184	471/5184

The combined test qualifies 100% of O1 boxes and 32.08% of ASU boxes, versus 0.984% and 9.086% under the whole-quadrant criterion alone. Pooling weights sixteen-user cases more heavily; averaging the sixteen case fractions instead gives 32.565% combined coverage in ASU. High received SNR makes O1's finite power rectangles certifiably concave despite large squared correlations. ASU includes weak and zero channels and leaves more boxes to generic subdivision. These observations describe the two channel populations and do not establish that every high-SNR box is concave.

Achieved rates are assessed separately from upper bounds. The median increase over the common grid is 0.04035% (mean 0.3460%, maximum 1.5693%). Only three cases exceed 1%: ASU with $(K, j) = (8, 1), (16, 0), (16, 2)$. The 1% cutoff is descriptive; we perform no formal rate-superiority or equivalence test. Every delivered lower is at least the independently checked grid lower.

For context, relative to the saved Full allocations at $K = 8$ and Grid+pricing allocations at $K = 16$, the median increase is 0.03555%, with only two cases exceeding 1% and a maximum of 1.1804%. These historical allocations were rechecked on identical inputs but have different computational allowances and

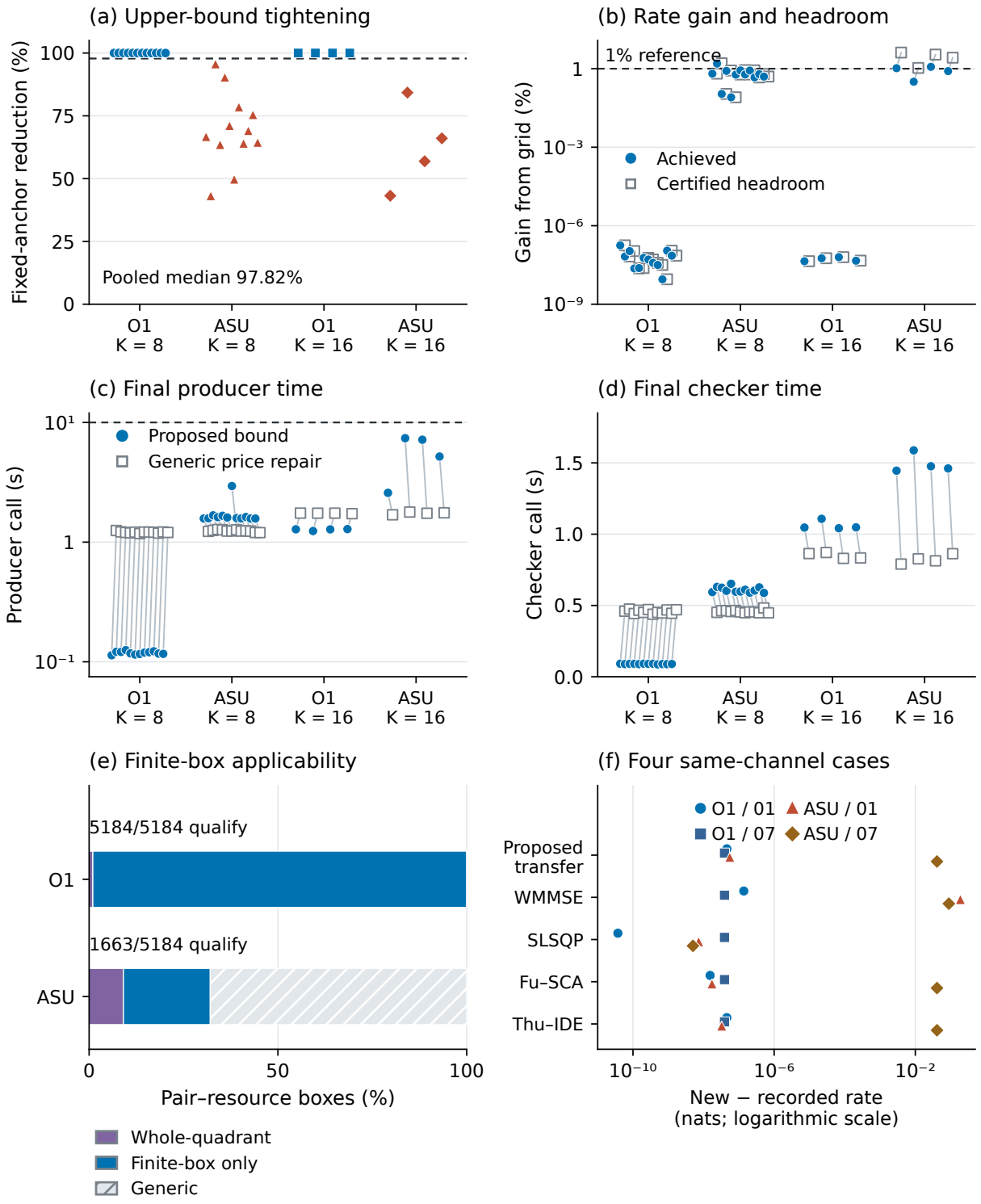


Figure S14: All-case diagnostics. (a) Fixed-anchor bound reduction. (b) Grid-relative achieved gains and certified headroom. (c)–(d) Callable construction/checking times. (e) Curvature classification. (f) Twenty historical refiner records on four identical physical inputs; case numbers are one-based. Timing scopes differ, and both WMMSE verification-timeout records retain valid feasible allocations. Fu-SCA and Thu-IDE use the component adaptations in [12, 13].

initialization/search scopes. The new uppers place every old O1 Full/pricing allocation within 1.382×10^{-7} nat of the continuous integral optimum. The corresponding upper on possible relative improvement is at most 0.725% across all eight-user cases and 4.143% across all 32 cases. Limited rate improvement is therefore consistent with limited available improvement, especially in O1.

We also re-evaluate the twenty saved T/S/W/Fu-SCA/Thu-IDE allocations on the four matching physical inputs. Their channels and common grid anchors agree exactly, but their timing scopes differ from this comparison. On those four inputs, the absolute achieved-rate differences between the new method and saved SLSQP allocations are at most 3.92×10^{-8} nat. These observed differences establish neither rate equivalence nor statistical superiority. The two recorded W verification timeouts, on ASU with $K = 8, j = 0$ and $j = 6$, remain associated with their valid retained feasible allocations.

S11.10 Interpretation of the bound improvement

The measured improvement combines finite-box curvature, changed power-price proposals and integral scheduling. The comparison in (B23) orders repaired dual and integral uppers only for the same λ, H; the two evaluated producers use different price proposals and pair-bound tables. Thus the experiment measures their combined effect rather than isolating any one component. All tables cover the full original power boxes, and the remaining gaps can reflect rectangular-bound error, the chosen prices and the budget relaxation.

In particular, the mixture argument in (B24) still applies: exact pair pricing and exact integral scheduling need not close the power-budget duality gap. The implementation uses numerical optimization to propose tangent points and powers, followed by rational enclosure checks; it does not integrate the algebraic quintic pricing oracle. The data support a substantial reduction in certified uncertainty and modest achieved-rate changes, not a broad rate-superiority conclusion. Results remain conditional on the two fixed ray-traced scenes and the nominal channel model; received SNR and useful-symbol rate are not deployment-throughput measurements.

S12 Receive-Dimension Benchmark and Exact-Search Details

This section documents the Phase-1 benchmark used in the receive-dimension and evaluation sections of the main paper. The original 32 stored DeepMIMO cases are transformed deterministically. No case is selected by the outcome of the proposed method.

S12.1 Construction

For each stored case, the channel vectors are truncated to the first M receive coordinates, with $M \in \{2, 4, 8\}$. The power box is $[0.001, 0.2]$ W and the shared budget is $P/K = 0.13$ W. The baseline grid is $\{0.05, 0.1, 0.2\}$ W. Every method starts from sequential pairing $(2s, 2s + 1)$ at uniform power P/K. The $K = 8$ runs use 12 s and the $K = 16$ runs 20 s; the controlled scale check uses 10 s.

S12.2 Pairing headroom

Table S18 reports the exact fixed-power pairing gain at uniform power. At $M = 8$ the median is zero; the same channels have 2.61% and 8.87% median gain at $M = 4$ and $M = 2$.

Table S18: Pairing gain at uniform power after deterministic M-truncation, all 32 stored cases.

M	Median	90th pct.	$> 1\%$	$> 5\%$
2	8.87%	21.56%	93.8%	68.8%
4	2.61%	8.07%	71.9%	0.0%
8	0.00%	0.25%	3.1%	0.0%

S12.3 Rate summary

Table S19 repeats the aggregate rates of the main paper and adds the number of cases. Table S20 repeats the paired differences. The absolute conversion uses $Q/(S \ln 2)$ with $S = 8$ for $K = 16$ and $S = 4$ for $K = 8$.

Table S19: Sum rate in nat, mean/median.

Population	Grid	SLSQP	I-WMMSE	Matching	Proposed
$K = 8$, ray	34.53/38.38	36.27/40.70	34.07/39.11	35.60/39.60	36.32/40.70
$K = 16$, ray	66.04/74.48	61.60/69.41	59.31/64.83	67.19/75.43	68.11/76.69
$K = 16$, ctl	46.36/40.93	43.40/39.26	40.79/36.82	47.00/41.76	47.36/41.91

Table S20: Proposed minus each common-model baseline at $M = 2$.

Population	Baseline	Median	Mean	95% CI	W/T/L
$K = 8$, ray	SLSQP	0.00%	0.08%	$[-0.15, 0.30]$%	5/17/2
$K = 8$, ray	Matching	2.06%	2.32%	$[1.50, 3.16]$%	21/3/0
$K = 16$, ray	Matching	1.37%	1.53%	$[0.98, 2.08]$%	8/0/0
$K = 16$, ctl	Matching	0.46%	0.64%	$[0.30, 1.10]$%	11/1/0

S12.4 Certification boundary at $M = 2$

The frozen finite-box/integral producer is run for 10 s on each proposed $M = 2$ schedule. Ten cases certify to a gap of about 10^{-12} nat; the remaining 22 cases have gaps between 0.25 and 30 nat at the 1200-split cap. The loose cases are concentrated in O1 and $K = 16$. The interval remains valid but does not support a certified rate claim.

S12.5 Algorithm details

The exact assignment recurrence is

$$V(A) = \max_{\{u,v\} \subseteq A} \left[V(A \setminus \{u, v\}) + w_{uv,\,|A|/2-1} \right], \tag{1}$$

with $V(\varnothing) = 0$ and $w_{uvs} = q_{r_{uvs}}(g_{us}p_u, g_{vs}p_v)$. It uses $O(2^K)$ states and $O(K^2)$ transitions per state. For $K = 16$ the state count is 65 536. For $K = 32$ the implementation falls back to matching.

At a fixed assignment, the power problem is solved by SQP with the analytic gradient of the pair rate. The repair step clips each power to the box and removes any excess budget from the largest slacks. The alternating search uses one deterministic and seven randomized starts and stops when the assignment and powers change by less than 10^{-6}.

S12.6 Scope

This section is a ray-traced cross-check, not the rate headline. The $K = 16$ ray-traced result rests on eight cases and is not used for the rate claim. The rate claim uses the 3GPP CDL benchmark in Section S13, which has 53 cases. The ray-traced result is retained because it explains the $M = 8$ null and supports the receive-dimension mechanism.

S13 3GPP CDL Benchmark

This section documents the standardized benchmark used for the rate claim in the main paper's evaluation section. The cluster delays, powers and angles are taken from the 3GPP TR 38.901 CDL-A, CDL-B and CDL-C tables. Each user receives a mean arrival angle; cluster angles are offset by the profile azimuth spread. A half-wavelength ULA with $M = 2$ coordinates is used, with independent per-cluster phases and a

100 ns delay spread. The eight resources are the first eight tones of a 50 MHz, 32-tone grid. The generator is seed-reproducible and included with the artifact.

S13.1 Population

The primary population has 53 cases with $K = 16$ and $S = 8$: three profiles, four angular separations, four seeds and five separation-sweep cases. The secondary population has 24 cases with $K = 8$ and $S = 4$. The power box is $[0.001, 0.2]$ W and the budget is $P/K = 0.13$ W. The grid is $\{0.05, 0.1, 0.2\}$ W. Every method starts from sequential pairing at uniform power P/K.

S13.2 Validation statistics

Table S21 gives the generated correlation and SNR ranges. The median squared correlation is about 0.5, the 90th percentile is about 0.92, and the median received SNR at 0.1 W is 12–14 dB. These are correlated, interference-limited channels, not the high-SNR nearly orthogonal regime of the earlier ray-traced benchmark.

Table S21: Generated CDL statistics, median over cases.

Profile	r median	r 90th pct.	SNR (dB)
CDL-A	0.46–0.56	0.91–0.94	12.2–13.9
CDL-B	0.49–0.52	0.90–0.92	12.2–13.9
CDL-C	0.47–0.55	0.90–0.93	12.2–13.9

S13.3 Rate summary

Table S22 gives the aggregate rates. Table S23 gives the paired contrasts. The $K = 16$ gain is 1.31% median with a 95% interval $[1.19, 1.42]$% and 53 wins in 53 cases. The absolute median gain is 0.129 bit/s/Hz. At $K = 8$ the gain over SLSQP is 0.044% median with a negligible absolute value.

Table S22: CDL sum rate in nat, mean/median.

Population	Grid	SLSQP	I-WMMSE	Matching	Proposed
$K = 16$	53.60/53.55	47.46/46.24	45.49/44.37	53.78/53.67	54.56/54.39
$K = 8$	25.87/26.82	26.45/27.18	24.56/23.71	26.34/27.17	26.52/27.41

Table S23: Proposed minus each common-model baseline on CDL.

Population	Baseline	Median	Mean	95% CI	W/T/L	n
$K = 16$	best	1.31%	1.31%	$[1.19, 1.42]$%	53/0/0	53
$K = 16$	matching	1.31%	1.31%	$[1.19, 1.42]$%	53/0/0	53
$K = 8$	SLSQP	0.044%	0.274%	$[0.106, 0.485]$%	13/10/1	24

S13.4 Attribution

The CDL ablation gives mean gains over the uniform anchor of 54.8% for assignment only, 17.3% for power only and 54.9% for the joint method. The exact assignment is the dominant mechanism; the SQP power step adds a small refinement. This is consistent with the model at $M = 2$: pairing is the main discrete lever. A 30-s sensitivity on 12 CDL cases gives matching–WMMSE and SLSQP more time. Matching–WMMSE improves, but the proposed 10-s result remains ahead in all 12 cases, with a median gain of 1.07%. This addresses the concern that the baseline field is weak because of the time limit.

S13.5 Relation to the ray-traced cross-check

The earlier DeepMIMO O1/ASU study is retained only as a ray-traced cross-check. At $M = 8$ it has zero fixed-power pairing gain in all 32 cases and a strong certificate (27/32 at 0.01 nat, 97.82% median gap reduction). The $K = 16$ DeepMIMO rate result is not used for the headline because it rests on eight cases. The CDL benchmark replaces it for the rate claim.

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